



AWS Solutions Architect Professional (SAP-C02)

300 Interview Q&A

- ☑ Comprehensive Interview Preparation Guide
- ☑ 30 Topics Covering All SAP-C02 Domains
- ☑ Detailed Answers with Best Practices
- ☑ Scenario-Based Questions & Architecture Tips

Akhilesh Chaudhari

Senior IT Engineer • akhileshchaudhari.com

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Q1

What are the five pillars of the AWS Well-Architected Framework?

▣ ANSWER

Operational Excellence, Security, Reliability, Performance Efficiency, and Cost Optimization. These pillars guide cloud architects in building secure, high-performing, resilient, and efficient infrastructure.

Q2

What is the difference between Availability Zones and Regions in AWS?

▣ ANSWER

A Region is a geographic area containing multiple isolated Availability Zones (AZs). Each AZ consists of one or more data centers with redundant power, networking, and connectivity. AZs within a region are interconnected via low-latency links.

Q3

Explain the concept of Shared Responsibility Model in AWS.

▣ ANSWER

AWS manages security 'of' the cloud (hardware, software, networking, facilities), while customers manage security 'in' the cloud (data, applications, access management, OS configurations, network/firewall settings).

Q4

What is AWS Organizations and how does it help enterprises?

▣ ANSWER

AWS Organizations lets you centrally manage multiple AWS accounts. It enables consolidated billing, Service Control Policies (SCPs), and organizational units (OUs) to enforce governance and compliance across accounts.

Q5

What is a Service Control Policy (SCP)?

▣ ANSWER

SCPs are permission policies applied at the AWS Organizations level. They set maximum permissions for accounts within an organization but do not grant permissions — they act as a guardrail filtering what IAM policies can allow.

Q6

How does AWS Control Tower simplify multi-account governance?

▣ ANSWER

AWS Control Tower automates the setup of a landing zone with best-practice guardrails, pre-configured accounts, centralized logging, and SSO integration. It uses Account Factory to provision new accounts with standardized configurations.

Q7

What is the difference between a vertical and horizontal scaling strategy?

▣ ANSWER

Vertical scaling (scale up) increases the capacity of a single resource (e.g., larger EC2 instance). Horizontal scaling (scale out) adds more instances to distribute load. Horizontal scaling is preferred in AWS for fault tolerance and elasticity.

Q8

Explain loose coupling in AWS architecture.

▣ ANSWER

Loose coupling means designing components that interact through well-defined interfaces (like SQS queues or API Gateway) rather than direct dependencies. This increases resilience — one component's failure doesn't cascade to others.

Q9

What is the principle of 'design for failure' in AWS?

▣ ANSWER

Assume every component will fail and architect accordingly. Use Auto Scaling, Multi-AZ deployments, Route 53 health checks, and cross-region replication to ensure applications continue functioning even when components fail.

Q10

When would you use AWS Outposts?

▣ ANSWER

AWS Outposts extends AWS infrastructure to on-premises data centers, providing the same AWS APIs, tools, and hardware. Use it when workloads require ultra-low latency, local data processing, or must remain on-premises due to regulatory requirements.

Section 2: Identity & Access Management (IAM)

Q11-Q20

Q11

What is the difference between IAM Roles and IAM Users?

▣ ANSWER

IAM Users have permanent long-term credentials (username/password or access keys). IAM Roles have temporary security credentials and are assumed by trusted entities (EC2, Lambda, cross-account users). Roles are preferred for applications and services.

Q12

How does IAM Policy evaluation work?

▣ ANSWER

AWS evaluates policies using these steps: explicit DENY wins over any ALLOW; if no explicit DENY, check for explicit ALLOW; if neither, default DENY applies. The evaluation considers identity-based policies, resource-based policies, SCPs, permission boundaries, and session policies.

Q13

What is a Permission Boundary in IAM?

▣ ANSWER

A permission boundary is an advanced IAM feature that sets the maximum permissions an IAM entity can have. Even if an identity-based policy grants more permissions, the effective permissions are the intersection of the identity policy and the permission boundary.

Q14

Explain the concept of IAM Access Analyzer.

▣ ANSWER

IAM Access Analyzer continuously monitors resource policies (S3, KMS, IAM roles, Lambda, SQS, Secrets Manager) to identify resources shared with external principals. It generates findings for resources accessible outside your AWS Organization or account.

Q15

What are IAM Identity Center (SSO) and when should you use it?

▣ ANSWER

AWS IAM Identity Center (formerly AWS SSO) provides centralized access management for multiple AWS accounts and applications. Use it in multi-account environments to provide employees single sign-on to all assigned accounts with permission sets.

Q16

How does cross-account IAM access work?

▣ ANSWER

Cross-account access involves: (1) Create an IAM role in the target account with a trust policy allowing the source account. (2) Attach permission policies to the role. (3) In the source account, create an IAM policy allowing sts:AssumeRole on the target role. Users/services then call sts:AssumeRole to get temporary credentials.

Q17

What is AWS STS and what are its key operations?

▣ ANSWER

AWS Security Token Service (STS) issues temporary, limited-privilege credentials. Key operations: AssumeRole (for role assumption), AssumeRoleWithSAML (for SAML federation), AssumeRoleWithWebIdentity (for OIDC/web identity federation), GetFederationToken, and GetSessionToken.

Q18

What are resource-based policies? Name AWS services that support them.

▣ ANSWER

Resource-based policies are attached directly to AWS resources (not IAM identities). Services that support them include: S3 (bucket policies), KMS (key policies), Lambda (function policies), SQS (queue policies), SNS (topic policies), ECR (repository policies), and Secrets Manager.

Q19

How does ABAC (Attribute-Based Access Control) work in AWS IAM?

▣ ANSWER

ABAC uses resource tags and IAM principal tags as policy conditions. For example, a developer can only manage EC2 instances tagged with Project:dev if both their IAM principal tag and resource tag match. ABAC scales well because new resources with matching tags automatically inherit correct permissions.

Q20

What is the difference between AWS Managed Policies and Customer Managed Policies?

▣ ANSWER

AWS Managed Policies are created and maintained by AWS, automatically updated as services evolve. Customer Managed Policies are created by you, offering more granular and customized control. Inline policies are embedded directly in a single user/role/group and are deleted when the entity is deleted.

Section 3: Compute – EC2, Auto Scaling & Lambda

Q21–Q30

Q21

What are the different EC2 instance purchasing options?

▣ ANSWER

On-Demand (pay per second/hour), Reserved Instances (1 or 3-year commitment, up to 72% savings), Savings Plans (flexible commitment, up to 66% savings), Spot Instances (up to 90% savings, interruptible), Dedicated Hosts (physical server, BYOL), and Dedicated Instances (hardware isolation).

Q22

How does EC2 Auto Scaling work?

▣ ANSWER

Auto Scaling monitors application health and automatically adjusts EC2 capacity. It uses Launch Templates, scaling policies (target tracking, step scaling, simple scaling, scheduled), and health checks (EC2 and ELB). The Auto Scaling Group maintains desired capacity within min/max bounds.

Q23

Explain the difference between scale-in and scale-out protection.

▣ ANSWER

Scale-out increases the number of instances; scale-in decreases them. Scale-in protection can be set at the ASG or instance level to prevent specific instances from termination during scale-in events, useful for long-running processes.

Q24

What is an EC2 Launch Template vs a Launch Configuration?

▣ ANSWER

Launch Templates are newer, support versioning, multiple instance types, Spot + On-Demand mix, T2/T3 unlimited, and are required for mixed instances policy. Launch Configurations are legacy, don't support versioning, and will eventually be deprecated.

Q25

What are EC2 Placement Groups?

▣ ANSWER

Cluster: All instances in one AZ on same rack for low latency, high throughput (HPC). Spread: Instances on different hardware reducing correlated failures (max 7 per AZ). Partition: Groups of instances isolated on different partitions (racks), used for HDFS, Cassandra, Kafka.

Q26

How does AWS Lambda handle concurrency?

▣ ANSWER

Lambda concurrency is the number of simultaneously executing function instances. Account-level concurrency limit is 1,000 by default. Reserved concurrency guarantees capacity for a function. Provisioned concurrency pre-initializes execution environments to reduce cold starts.

Q27

What is Lambda@Edge and CloudFront Functions? When to use each?

▣ ANSWER

Lambda@Edge runs Node.js/Python functions at CloudFront edge locations for request/response manipulation; supports complex logic, external network calls. CloudFront Functions runs JavaScript at 200+ PoPs for simple, high-frequency transformations like URL rewrites; sub-millisecond execution, 1/6 the cost.

Q28

Explain EC2 instance metadata and user data.

▣ ANSWER

Instance metadata (169.254.169.254/latest/meta-data/) provides data about the instance (instance ID, IP, IAM credentials). User data (169.254.169.254/latest/user-data/) runs scripts at launch. IMDSv2 uses session-oriented requests for enhanced security. Metadata is not encrypted in transit.

Q29

What is the difference between warm pools and on-demand instances in Auto Scaling?

▣ ANSWER

Warm pools maintain pre-initialized EC2 instances in a stopped or running state, dramatically reducing the time to scale out. Unlike on-demand cold launches, warm pool instances skip initialization steps and enter the InService state quickly.

Q30

When would you use AWS Batch over Lambda for compute workloads?

▣ ANSWER

AWS Batch is ideal for large-scale batch processing jobs requiring more than Lambda's 15-minute timeout, more than 10 GB memory, custom AMIs, or GPU instances. Lambda suits short, event-driven functions. Batch uses ECS/Fargate/EC2 and manages job queues, compute environments automatically.

Q31

What are the different S3 storage classes?

▣ ANSWER

S3 Standard (frequent access), S3 Intelligent-Tiering (auto-moves based on access), S3 Standard-IA (infrequent access), S3 One Zone-IA (single AZ), S3 Glacier Instant Retrieval, S3 Glacier Flexible Retrieval (minutes to hours), S3 Glacier Deep Archive (12+ hours), S3 Express One Zone (single-AZ, ultra-low latency).

Q32

How does S3 Versioning help with data protection?

▣ ANSWER

Versioning keeps multiple versions of an object, protecting against accidental deletions and overwrites. With versioning enabled, a delete adds a delete marker (doesn't remove data). Restoring means deleting the delete marker. Works with MFA Delete for additional protection.

Q33

What is S3 Replication and what are its types?

▣ ANSWER

S3 Replication automatically copies objects between buckets. Cross-Region Replication (CRR) replicates to different AWS regions for disaster recovery and compliance. Same-Region Replication (SRR) replicates within the same region for log aggregation and live replication between production and test accounts.

Q34

Explain S3 Object Lock and Glacier Vault Lock.

▣ ANSWER

S3 Object Lock prevents object deletion/overwrite using WORM (Write Once Read Many) model. Compliance mode: no one can override/delete (including root). Governance mode: users with special permission can. Glacier Vault Lock uses an immutable policy that locks after 24 hours; used for regulatory compliance.

Q35

What is the difference between EBS and EFS?

▣ ANSWER

EBS (Elastic Block Store): network-attached block storage, attached to one EC2 instance at a time (except multi-attach io1/io2), AZ-specific. EFS (Elastic File System): managed NFS file system, accessible from multiple EC2 instances across AZs simultaneously, automatically scales, POSIX-compliant.

Q36

What are EBS volume types and their use cases?

▣ ANSWER

gp3 (general purpose SSD, default): balanced price/performance. io2 Block Express (provisioned IOPS SSD): databases needing >16,000 IOPS. st1 (throughput HDD): big data, data warehouses, log processing. sc1 (cold HDD): infrequently accessed data. io1: legacy high-performance workloads.

Q37

What is S3 Transfer Acceleration?

▣ ANSWER

S3 Transfer Acceleration uses CloudFront's globally distributed edge locations to accelerate uploads to S3. Data is routed over optimized AWS network paths instead of the public internet. Useful for distant clients uploading large files. Costs extra per GB transferred.

Q38

How does S3 Lifecycle Policy work?

▣ ANSWER

S3 Lifecycle rules transition objects between storage classes or expire them after a set time. Example: move to Standard-IA after 30 days, Glacier after 90 days, delete after 365 days. Applies to current and non-current (versioned) objects separately.

Q39

Explain the difference between S3 server-side encryption options.

▣ ANSWER

SSE-S3: AWS manages keys (AES-256), free, default. SSE-KMS: customer controls keys via KMS, audit trail, extra cost. SSE-C: customer provides own keys (AWS does not store them). DSSE-KMS: dual-layer encryption using two KMS keys. Client-side encryption: encrypted before upload.

Q40

What is AWS Storage Gateway and its types?

▣ ANSWER

Storage Gateway connects on-premises apps to AWS storage. File Gateway: NFS/SMB interface backed by S3. Volume Gateway: iSCSI block storage (Stored: primary on-premises + async backup to S3; Cached: primary in S3 + frequently accessed cached locally). Tape Gateway: virtual tape library backed by S3/Glacier.

Section 5: Networking – VPC, Route 53 & CloudFront

Q41-Q50

Q41

What is a VPC and what are its key components?

▣ ANSWER

A Virtual Private Cloud (VPC) is an isolated virtual network in AWS. Key components: Subnets (public/private), Route Tables, Internet Gateway, NAT Gateway, Security Groups, Network ACLs, VPC Endpoints, Peering Connections, and Flow Logs.

Q42

What is the difference between Security Groups and NACLs?

▣ ANSWER

Security Groups are stateful (return traffic automatically allowed), operate at instance level, support only ALLOW rules. NACLs (Network ACLs) are stateless (return traffic must be explicitly allowed), operate at subnet level, support ALLOW and DENY rules, rules evaluated in number order.

Q43

Explain VPC Peering vs AWS Transit Gateway.

▣ ANSWER

VPC Peering: 1-to-1 private connection between two VPCs, no transitive routing, max 125 peering connections per VPC. Transit Gateway: hub-and-spoke model connecting thousands of VPCs and on-premises networks centrally, supports transitive routing, route tables, multicast, and inter-region peering.

Q44

What are VPC Endpoints and what types exist?

▣ ANSWER

VPC Endpoints allow private connectivity to AWS services without internet gateway or NAT. Interface Endpoints (PrivateLink): ENI with private IP, supports most AWS services, charged per hour and data. Gateway Endpoints: free, targets in route table, supports only S3 and DynamoDB.

Q45

How does AWS Direct Connect work?

▣ ANSWER

Direct Connect provides dedicated private network connections from on-premises to AWS (1 Gbps, 10 Gbps, 100 Gbps). Uses AWS Direct Connect locations and partner networks. Offers consistent network performance, reduced bandwidth costs. Use with VPN as backup for high availability.

Q46

What is the difference between Active-Active and Active-Passive HA in Route 53?

▣ ANSWER

Active-Active: all resources receive traffic simultaneously; if one fails, Route 53 stops routing to it. Active-Passive: primary resources serve traffic; secondary resources are standby. Route 53 routes to secondary only when primary health check fails.

Q47

Explain Route 53 Routing Policies.

▣ ANSWER

Simple (one record), Weighted (A/B testing, traffic distribution %), Latency-based (lowest network latency), Failover (active-passive), Geolocation (based on user location), Geoproximity (location + bias), Multivalue Answer (multiple healthy IPs, client-side load balancing), IP-based (CIDR-based routing).

Q48

What is AWS PrivateLink?

▣ ANSWER

AWS PrivateLink enables private connectivity between VPCs, AWS services, and on-premises applications using interface VPC endpoints without exposing traffic to the internet. Service providers create endpoint services; consumers create interface endpoints. Traffic stays on AWS network.

Q49

How does CloudFront handle caching and invalidation?

▣ ANSWER

CloudFront caches content at edge locations based on Cache-Control headers and TTL settings. Cache behaviors (path patterns) control caching per URL pattern. Invalidation removes objects from edge caches before TTL expires; first 1,000 invalidation paths/month are free. Use versioned file names instead of frequent invalidations.

Q50

What is the difference between NAT Gateway and NAT Instance?

▣ ANSWER

NAT Gateway: managed by AWS, highly available within an AZ, scales automatically up to 100 Gbps, no need to manage OS/patches, costs per hour + data. NAT Instance: EC2 instance running NAT, you manage it, can be used as bastion host, less costly but single point of failure, limited bandwidth.

Section 6: Databases – RDS, DynamoDB & ElastiCache

Q51-Q60

Q51

What is Amazon RDS Multi-AZ and how does it provide HA?

▣ ANSWER

RDS Multi-AZ maintains a synchronous standby replica in a different AZ. In case of planned/unplanned outage, RDS automatically fails over to the standby (typically 1-2 minutes). The CNAME record is updated to point to the standby. Used for high availability, not performance scaling.

Q52

What is the difference between RDS Read Replicas and Multi-AZ?

▣ ANSWER

Multi-AZ: synchronous replication, automatic failover, same region (cross-region option via automated backups), HA focus. Read Replicas: asynchronous replication, up to 15 replicas (Aurora), can be cross-region, used for read scaling, reporting, and can be promoted to standalone DB.

Q53

How does Amazon Aurora differ from standard RDS?

▣ ANSWER

Aurora is AWS-reengineered relational DB. Aurora stores data across 6 copies in 3 AZs automatically, tolerates 2-copy write and 3-copy read failures. Aurora Serverless v2 auto-scales in fractions of ACUs. 3x-5x faster than MySQL/PostgreSQL. Aurora Global Database spans multiple regions with sub-second replication.

Q54

What is DynamoDB and what are its core concepts?

▣ ANSWER

DynamoDB is a fully managed NoSQL key-value and document database. Core concepts: Tables, Items (rows), Attributes (columns), Primary Key (Partition Key or Partition+Sort Key), Secondary Indexes (GSI, LSI), Capacity Modes (Provisioned, On-Demand), DynamoDB Streams, and TTL.

Q55

What is the difference between DynamoDB GSI and LSI?

▣ ANSWER

LSI (Local Secondary Index): same partition key as base table, different sort key, must be created at table creation, 10 GB limit per partition key, uses base table capacity. GSI (Global Secondary Index): different partition and sort key, created anytime, own throughput capacity, no size limit.

Q56

How does DynamoDB DAX work?

▣ ANSWER

DAX (DynamoDB Accelerator) is an in-memory cache cluster for DynamoDB providing microsecond read performance. It's a write-through cache; writes go to DynamoDB immediately and to cache. Item cache (point reads) and query cache (query/scan results). No code changes needed — DAX is API-compatible.

Q57

What is DynamoDB Streams and when would you use it?

▣ ANSWER

DynamoDB Streams captures a time-ordered sequence of item-level changes in a table. Records are available for 24 hours. Use cases: trigger Lambda for event-driven architectures, cross-region replication, auditing, maintaining materialized views, analytics pipelines.

Q58

When would you use ElastiCache Redis vs Memcached?

▣ ANSWER

Redis: supports persistence, pub/sub, sorted sets, lists, replication, Multi-AZ, backup/restore, Lua scripting, streams, geospatial. Use for sessions, leaderboards, real-time analytics. Memcached: simpler, multithreaded, no persistence or replication. Use for simple distributed caching needing scale-out.

Q59

What is Amazon Redshift and when would you use it?

▣ ANSWER

Redshift is a fully managed petabyte-scale data warehouse using columnar storage and MPP (Massive Parallel Processing). Use for OLAP workloads, BI reporting, running complex SQL analytics on large datasets. Redshift Spectrum queries S3 data directly. RA3 nodes allow independent scaling of compute and storage.

Q60

How does Amazon Neptune work?

▣ ANSWER

Neptune is a fully managed graph database supporting Property Graph (Gremlin) and RDF (SPARQL). Use for social networks, recommendation engines, knowledge graphs, fraud detection, and network/IT operations. Neptune ML integrates with SageMaker for graph neural network-based ML predictions.

Section 7: Security & Compliance

Q61-Q70

Q61

What is AWS KMS and how does envelope encryption work?

▣ ANSWER

KMS manages cryptographic keys. Envelope encryption: data is encrypted with a data encryption key (DEK); the DEK is encrypted with a KMS CMK (Customer Master Key). Only the encrypted DEK is stored with the data. Decryption calls KMS to decrypt the DEK, then uses it to decrypt data. Reduces KMS API calls.

Q62

What is AWS CloudHSM and when would you use it over KMS?

▣ ANSWER

CloudHSM provides dedicated hardware security modules (FIPS 140-2 Level 3 certified). Use CloudHSM when you need full key custody (you manage keys; AWS has no access), specific hardware compliance requirements, or need to perform cryptographic operations within the HSM itself (TDE for Oracle, SSL offloading).

Q63

How does AWS WAF work?

▣ ANSWER

WAF is a web application firewall that filters HTTP/HTTPS traffic to ALB, API Gateway, CloudFront, and AppSync. Rules include IP sets, geo-match, rate-based rules, SQL injection/XSS filters, and managed rule groups. Web ACLs are created and associated with resources. Logs can go to S3, CloudWatch, or Kinesis Firehose.

Q64

What is AWS Shield and what are its tiers?

▣ ANSWER

Shield Standard: free, automatic protection against common network and transport layer DDoS attacks for all AWS customers. Shield Advanced: paid (\$3,000/month), 24/7 DDoS response team, cost protection, advanced attack visibility, integration with WAF, protection for EC2, ELB, CloudFront, Global Accelerator, Route 53.

Q65

How does AWS Macie protect sensitive data?

▣ ANSWER

Macie uses machine learning to automatically discover, classify, and protect sensitive data in S3. It identifies PII (names, credit cards, passport numbers), financial data, and custom patterns. Generates findings in Security Hub. Ideal for GDPR, HIPAA, and PCI-DSS compliance.

Q66

What is AWS GuardDuty?

▣ ANSWER

GuardDuty is a threat detection service that continuously analyzes CloudTrail, VPC Flow Logs, DNS logs, EKS audit logs, RDS login events, and S3 data events using ML and threat intelligence. Detects threats like cryptocurrency mining, unauthorized access, compromised instances, and exfiltration attempts.

Q67

Explain AWS Security Hub.

▣ ANSWER

Security Hub aggregates security findings from GuardDuty, Inspector, Macie, IAM Access Analyzer, Firewall Manager, and third-party tools into a single view. It runs automated compliance checks against CIS AWS Foundations, PCI DSS, and AWS Foundational Security Best Practices standards.

Q68

What is AWS Inspector and what does it scan?

▣ ANSWER

Inspector automatically discovers and scans EC2 instances, Lambda functions, and ECR container images for software vulnerabilities (CVEs) and unintended network exposure. Uses SSM Agent for EC2. Inspector v2 continuously monitors and provides a risk score based on CVSS score and exploitability.

Q69

How does AWS Certificate Manager (ACM) work?

▣ ANSWER

ACM provisions, manages, and auto-renews TLS/SSL certificates. Free for public certificates used with AWS services (ALB, CloudFront, API Gateway). Private certificates available via ACM Private CA (charged). Certificates cannot be exported for public CA; private CA certificates can be exported.

Q70

What is AWS Secrets Manager vs Parameter Store?

▣ ANSWER

Secrets Manager: specifically for secrets (DB passwords, API keys), automatic rotation via Lambda, cross-account sharing, ~\$0.40/secret/month. Parameter Store: general configuration data and secrets, free tier for standard parameters (max 4 KB), advanced parameters support 8 KB, policies, and higher throughput.

Section 8: Migration, Disaster Recovery & Business Continuity

Q71-Q80

Q71

What are the 7 Rs of cloud migration?

▣ ANSWER

Retire (decommission), Retain (keep on-premises), Rehost (lift-and-shift to EC2), Replatform (minor optimizations, e.g., RDS instead of self-managed DB), Repurchase (move to SaaS), Refactor/Re-architect (redesign using cloud-native services), Relocate (VMware Cloud on AWS, minimal effort).

Q72

What is the difference between RPO and RTO?

▣ ANSWER

RPO (Recovery Point Objective): maximum acceptable data loss, i.e., how old the data can be when recovered (e.g., RPO of 1 hour means data up to 1 hour old may be lost). RTO (Recovery Time Objective): maximum time to restore operations after a disaster. Lower RPO/RTO = higher cost.

Q73

Explain the four DR strategies with example architectures.

▣ ANSWER

Backup & Restore (highest RPO/RTO, lowest cost): S3 snapshots. Pilot Light (critical core always running): minimal EC2 + RDS, scale up on disaster. Warm Standby (scaled-down full system running): ALB + ASG + smaller DB, scale up. Multi-Site Active/Active (lowest RPO/RTO, highest cost): Route 53 failover across regions.

Q74

How does AWS Database Migration Service (DMS) work?

▣ ANSWER

DMS migrates databases to AWS with minimal downtime. Creates a replication instance, source/target endpoints, and migration tasks. Supports homogeneous migrations (MySQL to MySQL) and heterogeneous migrations (Oracle to Aurora) using Schema Conversion Tool (SCT). Supports ongoing replication via CDC.

Q75

What is AWS Application Migration Service (MGN)?

▣ ANSWER

MGN (formerly CloudEndure) performs lift-and-shift migrations by continuously replicating source servers to AWS using block-level replication. Creates a staging area in your target VPC. Cutover launches fully provisioned EC2 instances. Supports physical servers, VMware, Hyper-V, and cloud VMs.

Q76

How do you architect a multi-region active-active application?

▣ ANSWER

Use Route 53 latency-based or geolocation routing. Deploy same application stack in 2+ regions. Use DynamoDB Global Tables or Aurora Global Database for active-active data sync. Use S3 Cross-Region Replication for static assets. Use Global Accelerator for IP failover. Handle conflict resolution at the application layer.

Q77

What is AWS Backup?

▣ ANSWER

AWS Backup is a centralized service to automate and manage backups across AWS services (EC2, EBS, RDS, DynamoDB, EFS, FSx, Storage Gateway). Supports backup plans with schedules, retention policies, cross-region and cross-account backup. Vault Lock enforces WORM compliance for backups.

Q78

What is the AWS Snow family?

▣ ANSWER

Snowcone: smallest (2-8 TB), can be mailed or used with DataSync. Snowball Edge Storage Optimized (80 TB): large data transfers, edge compute. Snowball Edge Compute Optimized (42 TB): ML at edge. Snowmobile: exabyte-scale (100 PB), physical truck. Used when network bandwidth is insufficient or too costly.

Q79

Explain AWS DataSync.

▣ ANSWER

DataSync automates transferring data between on-premises storage (NFS, SMB, HDFS) and AWS (S3, EFS, FSx). Uses an agent on-premises. Handles encryption, integrity verification, and delta transfers. Can run on a scheduled or one-time basis. Much faster than open-source tools, uses parallel multi-threaded architecture.

Q80

What is AWS Transfer Family?

▣ ANSWER

Transfer Family provides managed SFTP, FTPS, FTP, and AS2 (AS2 for B2B EDI) servers backed by S3 or EFS. No infrastructure management needed. Supports custom identity providers (Cognito, Active Directory, LDAP). Use for migrating SFTP workflows to AWS without changing clients or processes.

Section 9: Analytics, ML & AI Services

Q81-Q90

Q81

What is Amazon Kinesis and what are its components?

▣ ANSWER

Kinesis is a platform for real-time data streaming. Kinesis Data Streams: capture and store streaming data, retention 1-365 days, custom processing. Kinesis Data Firehose: load streaming data to S3/Redshift/Elasticsearch/Splunk, fully managed, near-real-time. Kinesis Data Analytics: SQL/Apache Flink on streaming data. Kinesis Video Streams: video streaming.

Q82

Explain Amazon EMR and its use cases.

▣ ANSWER

EMR is a managed Hadoop framework running on EC2 (or EKS/Serverless). Supports Spark, Hive, HBase, Flink, Presto, Pig. Use for log analysis, ETL, machine learning, genomics, clickstream analysis. Data stored in S3 (EMRFS) for separation of compute and storage. Spot instances reduce cost significantly.

Q83

What is AWS Glue and how does it work?

▣ ANSWER

Glue is a serverless ETL service. Components: Data Catalog (metadata repository for all data sources), Crawlers (auto-discover and catalog schemas from S3, RDS, Redshift, DynamoDB), ETL Jobs (Spark/Python scripts), Glue Studio (visual ETL authoring), Elastic Views, and DataBrew (no-code data preparation).

Q84

When would you use Athena vs Redshift?

▣ ANSWER

Athena: serverless, pay-per-query, SQL on S3 data via Glue Catalog, ideal for ad-hoc queries on structured/semi-structured data without data loading. Redshift: provisioned/serverless data warehouse, best for complex queries on structured data, regular BI workloads, requiring high performance and complex joins.

Q85

What is Amazon SageMaker?

▣ ANSWER

SageMaker is an end-to-end ML platform. Key features: Studio (IDE for ML), Data Wrangler (data prep), Feature Store, Training (managed, distributed, spot), Experiments, Model Registry, Endpoints (real-time, serverless, async, batch transform), Pipelines (MLOps), Canvas (no-code ML), and Ground Truth (labeling).

Q86

What is Amazon OpenSearch Service?

▣ ANSWER

OpenSearch Service (formerly Elasticsearch Service) provides managed OpenSearch/Elasticsearch clusters. Use for log analytics, full-text search, application monitoring, clickstream analytics. Supports OpenSearch Dashboards (formerly Kibana). Integrates with Kinesis Firehose, CloudWatch Logs, DynamoDB, S3.

Q87

What is AWS Lake Formation?

▣ ANSWER

Lake Formation simplifies building, securing, and managing data lakes. It manages data catalog, security (column/row-level), data ingestion, and access control for S3. Built on top of Glue. Supports fine-grained Lake Formation permissions combined with IAM, enabling centralized data governance.

Q88

What is Amazon QuickSight?

▣ ANSWER

QuickSight is a serverless BI and visualization service. Supports SPICE (in-memory calculation engine) for fast query performance. ML Insights include anomaly detection and forecasting. Supports embedding dashboards in applications. Connects to S3, Athena, Redshift, RDS, Salesforce, and more. ML-powered narratives.

Q89

How does Amazon Rekognition work?

▣ ANSWER

Rekognition provides deep learning-based image and video analysis. Features: object/scene/activity detection, facial analysis and recognition, text in image, custom labels (train on your images), content moderation, personal protective equipment detection, and celebrity recognition. Streaming video analysis via Kinesis Video Streams.

Q90

What is Amazon Bedrock?

▣ ANSWER

Bedrock is a fully managed service providing access to foundation models (FMs) from AI21 Labs, Anthropic, Cohere, Meta, Mistral, Stability AI, and Amazon via a single API. Supports RAG via Knowledge Bases, Agents for multi-step workflows, model fine-tuning, and Guardrails for safety. Pay per API call.

Section 10: Cost Optimization & Management

Q91-Q100

Q91

What tools does AWS offer for cost management?

▣ ANSWER

AWS Cost Explorer (visualize, understand, and forecast costs), Budgets (set cost/usage alerts and actions), Cost and Usage Report (CUR, most granular data, exports to S3/Athena/QuickSight), Trusted Advisor (cost optimization checks), Compute Optimizer (right-sizing recommendations), Pricing Calculator (estimate costs before deployment).

Q92

What is the difference between AWS Savings Plans and Reserved Instances?

▣ ANSWER

Savings Plans: flexible 1 or 3-year commitment (\$/hour), apply automatically to any EC2/Lambda/Fargate usage. Compute Savings Plans (66% savings, any region/instance/OS), EC2 Instance Savings Plans (72% savings, specific family/region). Reserved Instances: specific instance type/region/OS/tenancy, less flexible but up to 72% savings.

Q93

How do you right-size EC2 instances?

▣ ANSWER

Use AWS Compute Optimizer (ML-based recommendations using 14 days of CloudWatch metrics), Trusted Advisor, or Cost Explorer right-sizing. Check CPU, memory, network utilization; downsize oversized instances, switch instance families, or use graviton/AMD for better price-performance. Also consider Spot for fault-tolerant workloads.

Q94

What is S3 Intelligent-Tiering and how does it save cost?

▣ ANSWER

S3 Intelligent-Tiering monitors object access patterns and moves objects between four access tiers: Frequent, Infrequent (30-day transition), Archive Instant (90-day), Archive and Deep Archive (opt-in). No retrieval fees, no minimum storage duration for Frequent/Infrequent. Small monthly monitoring fee per object.

Q95

How do you reduce data transfer costs in AWS?

▣ ANSWER

Use VPC Endpoints (S3/DynamoDB Gateway endpoints eliminate data transfer costs). Use same-region resources. Use CloudFront to cache and serve from edge (reduced origin egress). Use Direct Connect for predictable high-volume data transfer. Use S3 Transfer Acceleration only when needed. Data transfer into AWS is always free.

Q96

What is AWS Cost Anomaly Detection?

▣ ANSWER

Cost Anomaly Detection uses ML to continuously monitor costs and detect unusual spending patterns. Creates monitors at linked account, service, cost category, or cost allocation tag level. Sends alerts via SNS when anomalies are detected. You set threshold (percentage or dollar amount) for alert sensitivity.

Q97

How can tagging help with cost allocation?

▣ ANSWER

Tags (key-value pairs on resources) are activated in Cost Explorer as cost allocation tags. This enables filtering/grouping costs by project, team, environment, or application. AWS generates detailed cost reports with tags in CUR. Best practice: enforce tagging via SCPs, Config Rules, and Tag Policies in Organizations.

Q98

What are EC2 Spot Instances best practices for SAP-C02?

▣ ANSWER

Use Spot with Auto Scaling Groups (capacity-optimized or diversified allocation strategy across instance families/AZs). Enable Spot Instance Interruption Notices (2-minute warning via EC2 metadata). Use Spot Fleet for flexible workloads. Decouple state using SQS/DynamoDB. Ideal for batch, CI/CD, ML training, stateless web.

Q99

How do you optimize Lambda costs?

▣ ANSWER

Right-size memory (Lambda Power Tuning tool automates this). Reduce execution duration (optimize code, use connection reuse, avoid repeated KMS/SSM calls — use caching). Use Graviton2 (arm64) for 20% cost reduction. Use Provisioned Concurrency only when cold starts are truly a problem. Avoid unnecessary invocations.

Q100

What is AWS Trusted Advisor and what categories does it cover?

▣ ANSWER

Trusted Advisor provides real-time guidance across five categories: Cost Optimization (underutilized resources, idle load balancers), Performance (high-utilization instances, CloudFront optimization), Security (S3 bucket permissions, MFA, exposed access keys), Fault Tolerance (EBS snapshots, RDS Multi-AZ), and Service Limits.

Section 11: Containers & Serverless Architecture

Q101-Q110

Q101

What is the difference between ECS and EKS?

▣ ANSWER

ECS (Elastic Container Service): AWS-native container orchestration, simpler, uses Task Definitions and Services, tightly integrated with AWS. EKS (Elastic Kubernetes Service): managed Kubernetes control plane, portability, rich ecosystem, more complex. Both support EC2 and Fargate launch types. ECS is cheaper; EKS suits multi-cloud or Kubernetes-first teams.

Q102

What is AWS Fargate?

▣ ANSWER

Fargate is a serverless compute engine for containers. No EC2 instances to manage — you define CPU/memory at the task level; AWS provisions, scales, patches, and manages the underlying infrastructure. Works with both ECS and EKS. Pay per vCPU/memory per second. Slower start than EC2; higher per-unit cost.

Q103

Explain the ECS Task Definition.

▣ ANSWER

A Task Definition is a JSON blueprint for containers. Defines: container image, CPU/memory limits, port mappings, environment variables, secrets (Secrets Manager/Parameter Store), volumes (EFS, bind mounts), network mode (awsvpc, bridge, host), logging configuration (awslogs), and IAM task role.

Q104

What is AWS App Mesh?

▣ ANSWER

App Mesh is a service mesh using the Envoy proxy sidecar injected alongside each microservice. Provides traffic management (canary deployments, retries, timeouts), observability (distributed tracing via X-Ray, metrics to CloudWatch), and security (mTLS between services). Works with ECS, EKS, EC2, and Lambda.

Q105

How does AWS Lambda handle event-driven architectures?

▣ ANSWER

Lambda integrates with many AWS event sources: S3 (ObjectCreated), DynamoDB Streams, Kinesis, SQS, SNS, EventBridge, API Gateway, ALB, Cognito, IoT, CloudWatch Events. Each event source has different invocation models: synchronous (API GW, ALB), asynchronous (S3, SNS), and polling (SQS, Kinesis, DynamoDB Streams).

Q106

What is Amazon ECR?

▣ ANSWER

ECR (Elastic Container Registry) is a managed Docker container registry. Supports public and private repositories. Lifecycle policies auto-delete old images. Image scanning (basic and Inspector-based enhanced scanning) for CVEs. Cross-region and cross-account replication. Integrates with ECS, EKS, CodeBuild, CodePipeline.

Q107

What is AWS Step Functions?

▣ ANSWER

Step Functions orchestrates workflows using state machines (JSON-based Amazon States Language). Standard Workflows: 1-year duration, exactly-once execution, async. Express Workflows: 5-minute max, at-least-once, high-volume event processing. Integrates with 200+ AWS services using optimized integrations. Visual workflow debugging in console.

Q108

How does API Gateway work and what types does it support?

▣ ANSWER

API Gateway manages APIs at scale. REST API: full-featured, caching, request/response transformations, usage plans, API keys. HTTP API: lower latency, cheaper, OIDC/OAuth2, Lambda proxy, JWT authorization, no API keys. WebSocket API: two-way communication for real-time apps. All support Lambda, HTTP backends, and AWS service integrations.

Q109

What is the difference between SQS Standard and FIFO queues?

▣ ANSWER

Standard: unlimited throughput, at-least-once delivery, best-effort ordering, near-infinite message size. FIFO: exactly-once processing, strict ordering, 3,000 TPS with batching (300 TPS without), 10 message groups. FIFO costs more. Use Standard for high throughput; FIFO for order-critical workflows (payment processing, inventory).

Q110

What is Amazon EventBridge?

▣ ANSWER

EventBridge is a serverless event bus connecting AWS services, SaaS apps (Salesforce, Zendesk, PagerDuty), and custom apps. Event rules match events and route to targets (Lambda, SQS, SNS, Step Functions, Kinesis, API destinations). Supports event archives and replay. Schema registry auto-discovers event schemas. Replaces CloudWatch Events.

Section 12: Monitoring, Logging & DevOps

Q111-Q120

Q111

What is the difference between CloudWatch Metrics, Logs, and Alarms?

▣ ANSWER

Metrics: time-series numerical data from AWS services and custom applications (1-second to 1-month resolution). Logs: text-based log data in Log Groups/Streams, queried via Logs Insights (SQL-like). Alarms: monitor metrics over a period, transitions between OK/ALARM/INSUFFICIENT_DATA, triggers SNS/Auto Scaling/EC2 actions.

Q112

What is CloudTrail and what does it log?

▣ ANSWER

CloudTrail records API calls made to AWS services (management events by default). Includes: who made the call (identity), when, from where (IP), what resource, and what request/response. Data events (S3 object-level, Lambda invocations) are optional. Insights detects unusual API activity. Logs stored in S3, optional CloudWatch Logs.

Q113

Explain the AWS CI/CD pipeline using native services.

▣ ANSWER

CodeCommit (Git repository) → CodeBuild (managed build, unit tests, creates artifacts) → CodeDeploy (deploys to EC2/ECS/Lambda using deployment configurations: AllAtOnce, HalfAtATime, OneAtATime, Canary, Linear) → CodePipeline (orchestrates the stages). Can also use GitHub/Bitbucket with CodePipeline.

Q114

What is AWS CloudFormation and its key concepts?

▣ ANSWER

CloudFormation provisions and manages AWS resources via infrastructure-as-code (YAML/JSON templates). Key concepts: Stack (deployed resources), StackSets (multi-account/region deployments), Change Sets (preview changes before deployment), Drift Detection (detect manual changes), Nested Stacks, Dynamic References (Secrets Manager, Parameter Store).

Q115

What is AWS CDK?

▣ ANSWER

AWS CDK (Cloud Development Kit) lets you define cloud infrastructure using familiar programming languages (TypeScript, Python, Java, C#, Go). CDK Apps contain Stacks, which contain Constructs (L1: direct CloudFormation, L2: higher-level, L3: opinionated patterns). CDK synthesizes to CloudFormation templates. Allows loops, conditionals, OOP in IaC.

Q116

What is Amazon CloudWatch Container Insights?

▣ ANSWER

Container Insights collects, aggregates, and summarizes metrics and logs from ECS, EKS, and Kubernetes on EC2. Provides pod-level CPU, memory, network, and disk metrics. Uses a CloudWatch agent and Fluent Bit for log aggregation. Creates pre-built dashboards and CloudWatch alarms for container environments.

Q117

What is AWS X-Ray?

▣ ANSWER

X-Ray provides distributed tracing for applications. Traces requests as they travel through services, capturing latency, status, and metadata. Service Map visualizes the application. Groups and filter expressions identify problematic traces. Integrates with API Gateway, Lambda, ECS, EC2, Elastic Beanstalk. SDK available for Java, Python, Node.js, Go, Ruby, .NET.

Q118

Explain AWS Config and its use cases.

▣ ANSWER

AWS Config continuously records configuration changes to AWS resources. Config Rules evaluate resources against desired configurations (e.g., all S3 buckets must have encryption enabled). Conformance Packs group rules for compliance frameworks (PCI, HIPAA). Remediation actions (manual or automatic via SSM Automation) fix non-compliant resources.

Q119

What is Amazon Managed Grafana and when would you use it?

▣ ANSWER

Managed Grafana is a fully managed data visualization service. Integrates natively with CloudWatch, Prometheus, X-Ray, Timestream, OpenSearch, Redshift, and Athena data sources. Use for unified observability dashboards across multiple data sources. Supports SAML/OIDC SSO, fine-grained permissions via IAM Identity Center.

Q120

What is AWS Systems Manager and its key features for operations?

▣ ANSWER

SSM is an operations hub for managing AWS and on-premises infrastructure. Key features: Session Manager (browser/CLI shell without SSH/RDP), Patch Manager (automated OS patching), Run Command (remote execution), State Manager (desired state), Parameter Store, Inventory, OpsCenter, Change Manager, and Automation (runbooks).

Section 13: Advanced Networking & Hybrid Cloud

Q121-Q130

Q121

What is AWS Global Accelerator and how is it different from CloudFront?

▣ ANSWER

Global Accelerator uses AWS's global network to improve availability and performance of TCP/UDP applications using static Anycast IPs. Best for non-HTTP use cases (gaming, IoT, VoIP), regional failover, and IP whitelisting. CloudFront is CDN for caching HTTP/HTTPS content at edge locations. GA routes to nearest healthy endpoint; CF serves cached content.

Q122

How does AWS Site-to-Site VPN work?

▣ ANSWER

Site-to-Site VPN creates an IPsec encrypted tunnel between on-premises network and AWS VPC via a Virtual Private Gateway (VGW) or Transit Gateway. Two tunnels for HA (different AWS endpoints). Supports BGP (dynamic routing) or static routing. Bandwidth limited to ~1.25 Gbps. Can be used as backup for Direct Connect.

Q123

What is SD-WAN integration with AWS and why does it matter?

▣ ANSWER

SD-WAN solutions from partners (Cisco Meraki, VMware VeloCloud) can connect directly to Transit Gateway via Connect attachment (GRE tunnel + BGP) providing higher performance (up to 5 Gbps per ECMP) and easier branch-to-cloud connectivity without per-branch VPN tunnels.

Q124

Explain AWS Network Firewall.

▣ ANSWER

Network Firewall is a managed stateful/stateless firewall deployed in a VPC subnet. Uses rules: stateless (5-tuple ACL-like), stateful (Suricata-based IPS/IDS rules), domain lists (allow/deny by domain), TLS inspection. Scales automatically. Centralize in a dedicated inspection VPC with Transit Gateway for multi-VPC architectures.

Q125

What is the difference between Direct Connect Dedicated and Hosted Connections?

▣ ANSWER

Dedicated: physical Ethernet port on AWS network device (1G, 10G, 100G) ordered directly through AWS, requires LOA-CFA and cross-connect at DX location. Hosted: smaller capacities (50 Mbps – 10 Gbps) provisioned through APN partners, faster to establish, shared physical port, bandwidth can be elastic.

Q126

How do you design a hub-and-spoke network topology on AWS?

▣ ANSWER

Deploy Transit Gateway as the hub. Attach all VPCs (spokes), Direct Connect Gateways, and VPN connections to TGW. Create TGW route tables to control routing between spokes (full mesh or isolated). Use TGW Network Manager for global visibility. Centralize egress (internet) and inspection (Network Firewall) in a dedicated VPC.

Q127

What is Amazon VPC IP Address Manager (IPAM)?

▣ ANSWER

IPAM automates IP address planning, tracking, and monitoring across AWS Regions and accounts. Integrates with AWS Organizations. Allocates CIDR ranges from pools, monitors IP utilization, prevents overlapping allocations, and enables automated VPC provisioning within defined IP spaces.

Q128

What is AWS Cloud WAN?

▣ ANSWER

Cloud WAN builds, manages, and monitors a global network connecting AWS VPCs and on-premises locations from a central console. Uses a Core Network (global) with Segments for routing domains, Edge Locations in selected regions, and Network Policies (JSON). Replaces manual Transit Gateway peering across regions.

Q129

How does AWS Wavelength work?

▣ ANSWER

Wavelength embeds AWS compute and storage at the edge of 5G networks (in telecom carrier data centers). Applications deployed in Wavelength Zones can serve mobile/5G users with ultra-low latency (<10 ms) without leaving the carrier network. Seamlessly connected back to AWS Region for full AWS service access.

Q130

What are the key considerations for architecting multi-account network security?

▣ ANSWER

Centralize inspection with AWS Network Firewall or third-party appliances in a shared services VPC. Use Transit Gateway for connectivity with route tables enforcing traffic flows. Deploy AWS Firewall Manager to centrally configure WAF, Shield, Security Groups, and Network Firewall across accounts. Use VPC Flow Logs and GuardDuty for detection.

Q131

A company has a 3-tier app needing HA. How do you architect it on AWS?

▣ ANSWER

Web tier: ALB + Auto Scaling Group of EC2 across ≥ 2 AZs. Application tier: internal ALB + ASG across ≥ 2 AZs in private subnets. Database tier: RDS Multi-AZ (primary + standby in different AZs). Use Route 53 failover for regional DR. Store sessions in ElastiCache Redis. Use CloudFront for static assets. Enable ALB access logs.

Q132

How would you migrate a large Oracle database to AWS with minimal downtime?

▣ ANSWER

Use AWS DMS with Schema Conversion Tool (SCT) to convert Oracle to Aurora PostgreSQL. Set up DMS replication instance and full-load task. After full load, enable CDC (Change Data Capture) for ongoing replication. Test application against Aurora. During cutover window, stop writes to Oracle, let DMS sync final changes, redirect application to Aurora.

Q133

A company needs to process 1 million messages per day with guaranteed order. Which service?

▣ ANSWER

Use SQS FIFO queues with Message Group ID to maintain order within groups. For 1M messages/day (~ 12 messages/second), FIFO's 3,000 TPS (with batching) is more than sufficient. Lambda with SQS event source mapping processes messages. Use DLQ for failed messages. Consider SNS FIFO + SQS FIFO fanout if multiple consumers needed.

Q134

How do you reduce costs for a batch processing workload running 8 hours daily?

▣ ANSWER

Use EC2 Spot Instances with Auto Scaling Group (diversified across instance types/AZs). Save state to S3/DynamoDB for Spot interruption tolerance. If using EMR, use Spot for task nodes, On-Demand for core nodes. Use Savings Plans for the baseline On-Demand usage. Evaluate Fargate Spot for container workloads. AWS Batch manages job queue and compute.

Q135

A company requires all API calls to be logged and audited. How?

▣ ANSWER

Enable CloudTrail in all regions with a multi-region trail. Enable CloudTrail Insights for anomaly detection. Send logs to a centralized S3 bucket in a dedicated logging account (use S3 Object Lock for immutability). Enable log file validation. For real-time analysis, stream to CloudWatch Logs and create metric filters/alarms. Use Athena + QuickSight for audit reports.

Q136

How do you implement blue/green deployments on AWS?

▣ ANSWER

Option 1 (DNS-based): Two identical environments, Route 53 weighted routing shifts traffic gradually. Option 2 (ALB): Swap target groups or use weighted target group routing. Option 3 (ECS): CodeDeploy Blue/Green replaces old task set with new, validates, then terminates old. Option 4 (Lambda): Deploy new version, shift traffic using weighted aliases (canary/linear deployment configuration).

Q137

A company needs to enforce encryption at rest for all new S3 buckets in the organization. How?

▣ ANSWER

Use AWS Organizations Service Control Policy to deny s3:CreateBucket if aws:RequestedRegion doesn't have encryption enabled. Alternatively, use AWS Config Rule 's3-bucket-server-side-encryption-enabled' with automatic remediation via SSM Automation to enable SSE-S3 or SSE-KMS. Use CloudFormation StackSets with Bucket policies and encryption settings.

Q138

How do you architect a serverless data processing pipeline for real-time analytics?

▣ ANSWER

Data sources → Kinesis Data Streams (real-time ingestion) → Lambda (transform/enrich) → Kinesis Firehose (buffer, compress, encrypt) → S3 (data lake). Alternatively: Kinesis Data Analytics (Apache Flink) for real-time aggregations → DynamoDB or OpenSearch for low-latency queries. Glue/Athena for ad-hoc analysis. QuickSight for dashboards.

Q139

What does it mean to 'decouple' a monolithic application in AWS?

▣ ANSWER

Extract services and use SQS between them instead of direct synchronous calls. Use SNS fan-out for event distribution. Replace shared databases with service-specific databases. Use API Gateway for external interfaces. Introduce Strangler Fig pattern: gradually replace monolith routes with new microservices. Use Step Functions for workflow orchestration.

Q140

A global company needs 99.999% uptime. How do you design their architecture?

▣ ANSWER

Multi-region active-active deployment. Route 53 latency-based routing with health checks. Aurora Global Database with read replicas in each region. DynamoDB Global Tables for session/state. CloudFront with multiple origins. S3 Cross-Region Replication. Global Accelerator for static IPs. Automated failover testing via GameDay exercises. Chaos engineering with AWS Fault Injection Simulator.

Section 15: Emerging Services & Additional Topics

Q141-Q150

Q141

What is AWS Resilience Hub?

▣ ANSWER

Resilience Hub assesses and improves the resilience of applications by defining an RTO/RPO target and comparing the current architecture against that target. It analyzes CloudFormation stacks, Terraform state, or App Registry, identifies gaps, suggests improvements, and runs validation tests to confirm resilience posture.

Q142

What is AWS Fault Injection Simulator (FIS)?

▣ ANSWER

FIS is a managed chaos engineering service. Run controlled experiments to test application resilience by injecting faults (CPU stress, network latency, AZ failure, Spot interruption, RDS failover). Uses experiment templates defining targets, actions, and stop conditions. Integrates with CloudWatch for observability during experiments.

Q143

What is Amazon Connect and its cloud contact center capabilities?

▣ ANSWER

Connect is a cloud contact center service. Features: virtual contact center in minutes, AI-powered chatbots (Amazon Lex), real-time and historical analytics, agent workspace, customer profiles, wisdom (in-call knowledge articles), contact flows (IVR drag-and-drop), pay per minute (no hardware, no per-seat licensing), integrates with CRM via Lambda.

Q144

What is AWS AppSync?

▣ ANSWER

AppSync is a managed GraphQL API service. Supports real-time subscriptions via WebSocket, offline sync, and conflict resolution for mobile apps. Resolvers connect to DynamoDB, Lambda, RDS, HTTP endpoints, and OpenSearch. Integrates with Cognito for authentication. Supports merged APIs (federated GraphQL) for microservices.

Q145

What is Amazon Kendra?

▣ ANSWER

Kendra is an intelligent enterprise search service powered by ML. Understands natural language questions and returns specific answers (not just links). Connects to 40+ data sources (S3, SharePoint, Confluence, ServiceNow, Salesforce, RDS). Learns from feedback (incremental learning). Use for internal knowledge bases, FAQ search, document search.

Q146

What is AWS Elemental Media Services?

▣ ANSWER

MediaConvert: file-based video transcoding. MediaLive: live video encoding. MediaPackage: video origination and packaging (HLS, DASH, CMAF). MediaStore: low-latency media storage. MediaTailor: server-side ad insertion. Together they form a complete live and on-demand video streaming pipeline on AWS.

Q147

What is Amazon Managed Streaming for Apache Kafka (MSK)?

▣ ANSWER

MSK is a fully managed Kafka service. Provisions, operates, and maintains Kafka brokers and ZooKeeper nodes. Supports Kafka Connect (managed connectors to S3, DynamoDB, RDS) and MSK Serverless (auto-scales capacity). Integrates with Lambda as event source. Compatible with standard Kafka clients. Use for event streaming, data pipelines, log aggregation.

Q148

What is AWS AppConfig?

▣ ANSWER

AppConfig (part of AWS Systems Manager) manages application configuration separate from code. Deploys configuration changes with deployment strategies (linear, exponential rollout), validators (JSON schema, Lambda), and automatic rollback if CloudWatch alarms fire. Use for feature flags, throttle values, allow/deny lists without redeployment.

Q149

What is Amazon Location Service?

▣ ANSWER

Location Service provides geospatial data: maps (HERE, Esri), places/geocoding, routing, tracking (device movements), geofencing (proximity notifications when devices enter/exit), and geospatial queries. No per-user licensing. Data stays in AWS (not shared with map providers). Use for fleet tracking, delivery apps, asset management.

Q150

What is AWS Managed Service for Prometheus?

▣ ANSWER

Managed Prometheus is a serverless, Prometheus-compatible monitoring service. Automatically scales ingestion, storage, and querying. Integrates with Amazon Managed Grafana for visualization. Uses PromQL for querying. Works with EKS, ECS, EC2, and Lambda. No Prometheus server infrastructure to manage. Supports cross-region/cross-account metric collection.

Q151

What is a defense-in-depth strategy on AWS?

▣ ANSWER

Layered security using: edge protection (WAF, Shield, CloudFront), network isolation (VPC, NACLs, SGs), identity controls (IAM, MFA, SCPs), data protection (KMS, TLS, Macie), detection (GuardDuty, Inspector, Security Hub, CloudTrail), and incident response (automated remediation via Lambda + EventBridge).

Q152

How do you centralize security findings across multiple AWS accounts?

▣ ANSWER

Enable AWS Security Hub in all accounts with a delegated administrator account (or management account). Create an AWS Organizations integration to auto-enable Security Hub in new accounts. All findings from GuardDuty, Inspector, Macie, IAM Access Analyzer, and third-party tools aggregate in the administrator account.

Q153

What is AWS Firewall Manager?

▣ ANSWER

Firewall Manager centrally configures and manages WAF rules, Shield Advanced protections, VPC Security Groups, Network Firewall policies, and DNS Firewall rules across all accounts in an AWS Organization. Requires AWS Organizations and Security Hub. New accounts automatically inherit policies.

Q154

How do you implement zero-trust network access on AWS?

▣ ANSWER

Remove broad network access rules; verify every request. Use IAM for all access with least privilege. Use VPC endpoints to remove public access. Enforce mTLS with App Mesh. Use Verified Access (browser-based workload access without VPN, validates identity via IdP). Use PrivateLink for inter-service communication.

Q155

What is AWS Verified Access?

▣ ANSWER

Verified Access provides secure access to corporate applications without a VPN. It evaluates each request based on user identity (via IAM Identity Center, Okta, Ping) and device posture. Creates trust contexts per application policy. No client changes needed for browser-based apps. Logs access in CloudWatch, S3, or Kinesis Firehose.

Q156

How does Amazon Cognito work for application authentication?

▣ ANSWER

Cognito provides two main components: User Pools (manages user identity, handles sign-up/sign-in, MFA, hosted UI, JWT tokens) and Identity Pools (provides temporary AWS credentials to authenticated users via STS AssumeRoleWithWebIdentity). Can use external IdPs (Google, Facebook, SAML, OIDC) with User Pools.

Q157

What is SAML 2.0 federation with AWS?

▣ ANSWER

SAML 2.0 federation lets existing enterprise identity providers (AD FS, Okta, Ping) authenticate users who then assume IAM roles in AWS without creating IAM users. Ideal for: AWS Management Console access and programmatic API access. AWS supports SP-initiated and IdP-initiated SSO flows.

Q158

Explain AWS Network Access Control best practices.

▣ ANSWER

Principle of least privilege for SGs (specific port ranges, source CIDRs/SG references). Use NACLs as subnet-level defense layer. Enable VPC Flow Logs for traffic analysis. Use Reachability Analyzer to verify network paths. Regularly audit with IAM Access Analyzer for network exposure. Remove unused SG rules with Config.

Q159

What is AWS Detective?

▣ ANSWER

Detective automatically collects log data (CloudTrail, VPC Flow Logs, GuardDuty findings, EKS audit logs) and uses ML, statistical analysis, and graph theory to build interactive visualizations for security investigations. Helps analysts understand, investigate, and quickly identify root cause of security findings.

Q160

How do you protect against data exfiltration on AWS?

▣ ANSWER

Enable Macie (sensitive data discovery in S3), GuardDuty (unusual API calls, DNS queries to C2 servers), VPC Flow Logs (unusual traffic patterns), CloudTrail (data event logging for S3/Lambda). Use S3 bucket policies to restrict access. Block public S3 access at account level. Use PrivateLink to prevent traffic leaving AWS network.

Section 17: Well-Architected Framework Deep Dive

Q161-Q170

Q161

How do you apply the Operational Excellence pillar in practice?

▣ ANSWER

Define runbooks and playbooks in SSM Automation. Use Infrastructure as Code (CloudFormation/CDK). Implement observability (CloudWatch dashboards, X-Ray, Container Insights). Use CI/CD pipelines (CodePipeline). Conduct regular game days with FIS. Collect feedback via CloudWatch Logs Insights. Review OpsCenter items regularly.

Q162

What does the Reliability pillar require?

▣ ANSWER

Automatic recovery from failure (ASG health checks, RDS Multi-AZ). Horizontal scaling to prevent bottlenecks. Stop guessing capacity (Auto Scaling, Lambda). Manage change through automation (CloudFormation). Test recovery procedures (backups, DR failover). Use Route 53 health checks and failover routing.

Q163

How do you achieve the Security pillar's requirements?

▣ ANSWER

Implement strong identity foundation (MFA, least privilege, no long-term credentials). Enable traceability (CloudTrail, Config). Apply security at all layers (WAF, SGs, NACLs, ACM, KMS). Automate security best practices (Config Rules, Security Hub). Protect data in transit and at rest. Prepare for security events with incident response plans.

Q164

What are the key practices for Performance Efficiency pillar?

▣ ANSWER

Choose the right resource types/sizes (Compute Optimizer, Lambda Power Tuning). Use serverless where possible. Use caching (ElastiCache, CloudFront, DAX). Monitor performance metrics and set alarms. Go global in minutes using AWS Regions and edge services. Use benchmarks and load testing before production.

Q165

How do you implement the Cost Optimization pillar?

▣ ANSWER

Implement cloud financial management (tagging, budgets, CUR analysis). Right-size resources monthly using Compute Optimizer. Use pricing models appropriate to workload (Spot, Savings Plans, RI). Delete unused resources (EBS snapshots, unattached volumes, idle Load Balancers). Use managed services to reduce operational overhead.

Q166

What is the Sustainability pillar in AWS Well-Architected Framework?

▣ ANSWER

Sustainability (added in 2021) focuses on minimizing environmental impact. Practices: understand your impact (Customer Carbon Footprint Tool), establish sustainability goals, maximize utilization (right-sizing, Graviton), anticipate and adopt new efficient hardware, use managed services (AWS runs more efficiently at scale), reduce downstream impact.

Q167

What is a Well-Architected Review?

▣ ANSWER

A structured review using the AWS Well-Architected Tool to evaluate workloads against best practices. Answers milestone questions across the six pillars. Identifies High Risk Issues (HRIs) and Medium Risk Issues (MRIs). Generates improvement plan with prioritized recommendations. APN Partners can conduct reviews with IEM programs.

Q168

What are Well-Architected Lenses?

▣ ANSWER

Lenses extend the framework with domain-specific best practices. Official lenses include: Serverless, SaaS, Games, SAP, Machine Learning, Analytics, IoT, Healthcare, Financial Services, and HPC. Each adds lens-specific questions beyond the core framework pillars.

Q169

How does AWS Solutions Library help architects?

▣ ANSWER

Solutions Library provides vetted, technical reference implementations for common use cases (data lake, landing zone, IoT analytics, video on demand). Each solution includes documentation, CloudFormation templates, and customization guidance. Reduces time to implement common architectural patterns correctly.

Q170

What is the AWS Architecture Center?

▣ ANSWER

Architecture Center is a central repository of best practices, reference architectures, and whitepapers from AWS. Includes Well-Architected guidance, service-specific whitepapers, industry-specific reference architectures (e-commerce, genomics, financial), and prescriptive guidance for common scenarios.

Section 18: Data Lake & Modern Data Architecture

Q171-Q180

Q171

What are the key components of a modern data lake on AWS?

▣ ANSWER

Ingestion (Kinesis, Glue, DMS, DataSync, Transfer Family), Storage (S3 with intelligent tiering, organized in raw/clean/curated/enriched zones), Cataloging (Glue Data Catalog), Processing (EMR, Glue ETL, Lambda), Query (Athena), Governance (Lake Formation, Macie), Visualization (QuickSight), and Orchestration (Step Functions, Glue Workflows, MWAA).

Q172

What is Apache Iceberg and why is it important for data lakes on AWS?

▣ ANSWER

Iceberg is an open table format providing ACID transactions, schema evolution, time travel, and partition evolution for data lake tables. AWS Glue, Athena, and EMR natively support Iceberg. Enables UPDATE/DELETE on S3 data (unlike plain Parquet). S3 Tables is a purpose-built store for Iceberg tables with automatic compaction.

Q173

What is AWS Clean Rooms?

▣ ANSWER

Clean Rooms lets companies collaborate on combined datasets without sharing underlying data. Multiple parties query joined data within a clean room; results are protected by analysis rules (aggregation only, no row-level data revealed). Useful for advertising measurement, financial analysis, healthcare research with privacy compliance.

Q174

What is Amazon DataZone?

▣ ANSWER

DataZone is a data management service for cataloging, discovering, sharing, and governing data across organizational boundaries. Creates a business data catalog, manages subscriptions to data assets, enforces governance policies, and integrates with Glue, Redshift, Athena, and SageMaker for end-to-end data mesh architectures.

Q175

What is a data mesh architecture and how does AWS support it?

▣ ANSWER

Data mesh decentralizes ownership to domain teams (producers) who publish data products. Consumers discover and subscribe to data products. AWS supports this with: Lake Formation for governance, Glue for catalog, DataZone for marketplace, S3 for domain-owned storage, Athena/Redshift for consumption, and EventBridge for domain events.

Q176

Explain the medallion architecture (bronze/silver/gold) on AWS.

▣ ANSWER

Bronze (raw): raw, unprocessed data landed from source systems in S3. Silver (clean): deduplicated, validated, standardized data, often in Iceberg/Parquet. Gold (curated): aggregated, business-ready data for specific use cases (Redshift tables, QuickSight SPICE). Glue or EMR Spark processes each layer; Lake Formation governs access per layer.

Q177

What is Amazon Redshift Spectrum?

▣ ANSWER

Redshift Spectrum enables running SQL queries directly against exabytes of data in S3 without loading into Redshift. Uses Redshift's query engine with massively parallel processing against external tables defined in Glue Data Catalog. Queries can join Redshift tables with S3 data. Pay per query (per TB scanned in S3).

Q178

How does AWS Glue DataBrew work?

▣ ANSWER

DataBrew is a visual, no-code data preparation tool. Data scientists and analysts can clean and normalize data with 250+ pre-built transformations. Profiles datasets to understand quality issues. Creates recipes (transformation steps) that can be applied to new data. Outputs to S3, Redshift, or RDS. Integrates with Glue workflows.

Q179

What is Apache Kafka vs Amazon Kinesis — when to use each?

▣ ANSWER

Kinesis: fully managed, native AWS integration, pay-per-shard/record, simpler setup, limited to 7-day retention, up to 1 MB records. Kafka (MSK): more ecosystem (connectors, Kafka Streams, KSQL), unlimited retention, larger records, complex setup, open-source portability. Use Kinesis for AWS-first simplicity; MSK for Kafka compatibility requirements.

Q180

What is AWS Entity Resolution?

▣ ANSWER

Entity Resolution matches and links related records across datasets using ML-based matching (no shared identifiers needed) or rule-based matching. Helps businesses understand customer 360, de-duplicate records, link first/third-party data without sharing raw data. Integrates with S3, Glue, and Clean Rooms.

Section 19: Application Integration Patterns

Q181-Q190

Q181

What is the Saga pattern and how is it implemented on AWS?

▣ ANSWER

Saga manages distributed transactions across microservices. Choreography-based: each service publishes events that trigger the next step (use EventBridge or SNS). Orchestration-based: central coordinator (Step Functions) calls each service and handles compensating transactions on failure. Use DynamoDB for saga state tracking.

Q182

What is the Circuit Breaker pattern in AWS?

▣ ANSWER

Circuit Breaker prevents cascading failures by stopping calls to a failing service. When failures exceed a threshold, the circuit 'opens' and requests fail fast with fallback responses. In AWS: implement with Lambda + DynamoDB (track failure count), or use App Mesh Envoy proxy (native circuit breaking), or Route 53 health checks with failover.

Q183

How does the Outbox Pattern prevent dual-write problems in microservices?

▣ ANSWER

Instead of writing to DB and publishing an event in two separate operations (risky), write both the DB change and the event to an outbox table atomically in the same DB transaction. A separate process (DynamoDB Streams + Lambda, or Debezium with MSK) reads the outbox and publishes events reliably.

Q184

What is CQRS and how is it applied on AWS?

▣ ANSWER

CQRS (Command Query Responsibility Segregation) separates read and write models. Writes: go to primary store (DynamoDB, RDS), trigger events (DynamoDB Streams, EventBridge). Reads: events update read-optimized stores (Elasticsearch/OpenSearch, ElastiCache, read replicas). Query the read store for complex queries without impacting write performance.

Q185

What is Event Sourcing and what AWS services support it?

▣ ANSWER

Event Sourcing stores state as a sequence of events rather than current state. Event store: DynamoDB or Kinesis Data Streams. Snapshots: stored in S3 or DynamoDB for fast rebuilds. Event replaying: Lambda processes events to rebuild projections. DynamoDB Streams captures change events. Kinesis records events with configurable retention (up to 365 days).

Q186

Explain the Fan-out pattern using SNS and SQS.

▣ ANSWER

An SNS topic receives a message and fans it out to multiple SQS queues (each subscribed to the topic). Each queue feeds independent processing Lambda functions or applications. Benefits: decoupling, independent scaling, failure isolation. Use SNS filter policies to route specific messages to specific queues instead of all.

Q187

What is Amazon EventBridge Pipes?

▣ ANSWER

EventBridge Pipes creates point-to-point integrations between event producers (DynamoDB Streams, Kinesis, SQS, MSK, MQ) and targets (Lambda, Step Functions, Kinesis, SQS, HTTP endpoints) with built-in filtering, enrichment (Lambda/API GW/EventBridge/Step Functions), and transformation. Eliminates custom polling Lambda functions.

Q188

How does Amazon MQ differ from SQS and SNS?

▣ ANSWER

MQ is a managed message broker for Apache ActiveMQ and RabbitMQ. Use when migrating existing applications using JMS, AMQP, MQTT, OpenWire, STOMP protocols — no code changes needed. SQS/SNS are AWS-native and scalable to unlimited throughput but require SDK changes. MQ has limited scaling vs SQS.

Q189

What is AWS AppFlow?

▣ ANSWER

AppFlow transfers data between SaaS applications (Salesforce, SAP, Zendesk, ServiceNow, Slack, Google Analytics) and AWS services (S3, Redshift, EventBridge). Supports scheduled, event-triggered, or on-demand flows. Data transformation: filter, map, concatenate, mask. Encrypts data in transit and at rest. No servers to manage.

Q190

What is the Strangler Fig pattern for microservices migration?

▣ ANSWER

Strangler Fig gradually replaces a monolith by intercepting calls and routing them to new services. In AWS: place API Gateway or ALB in front of the monolith. Route specific paths (e.g., /orders) to new microservices; remaining paths still hit the monolith. Over time, strangle the monolith until all paths are migrated to new services.

Q191

What are the most commonly tested SAP-C02 topics?

▣ ANSWER

High-frequency topics: Migration strategies (6Rs/7Rs), DR strategies (RTO/RPO), Multi-account governance (Organizations, Control Tower, SCPs), Cost optimization (Savings Plans vs RIs, Spot), Networking (TGW vs VPC Peering, Direct Connect), Security (KMS, WAF, GuardDuty, IAM policies), and Serverless vs container architectures.

Q192

How do you approach 'MOST cost-effective' questions on SAP-C02?

▣ ANSWER

Always favor: Spot Instances for stateless/batch workloads, Savings Plans over RIs for flexibility, S3 over EBS for object storage, Lambda over EC2 for event-driven, Serverless (Fargate, Aurora Serverless) for variable loads, Gateway VPC Endpoints (free) over NAT Gateway for S3/DynamoDB, and managed services over self-managed.

Q193

How do you approach 'MOST operationally efficient' questions?

▣ ANSWER

Choose fully managed services over self-managed equivalents. Prefer: ECS/EKS Fargate over EC2 with Docker, RDS/Aurora over EC2 with DB, MSK over self-managed Kafka, ElastiCache over EC2 Redis, AWS Backup over custom scripts, SSM Session Manager over bastion hosts, CloudWatch over custom monitoring.

Q194

What are 'anti-patterns' that SAP-C02 questions test?

▣ ANSWER

Common anti-patterns: using NAT Instance instead of NAT Gateway, using SSH/RDP instead of Session Manager, long-term IAM user credentials instead of roles, storing secrets in environment variables instead of Secrets Manager, using EBS for shared storage instead of EFS, single-AZ RDS without Multi-AZ for production databases.

Q195

How should you handle 'high availability' vs 'disaster recovery' distinctions?

▣ ANSWER

HA is about eliminating single points of failure within a region (Multi-AZ, ASG, Multi-AZ RDS). DR is about recovering from regional failures (cross-region replication, Route 53 failover). Questions mentioning '99.99% uptime' usually need HA. Questions mentioning 'region failure', 'RPO', 'RTO' need DR strategies.

Q196

What is the recommended approach for SAP-C02 scenario-based questions?

▣ ANSWER

Follow this framework: (1) Identify the key constraint (cost, performance, compliance, operational overhead). (2) Eliminate obviously wrong answers. (3) Among remaining, choose the one that best satisfies the constraint. AWS prefers managed services, automation, least privilege, and Pay-as-you-go. Read each word carefully — 'existing VPC', 'minimal changes', 'no downtime' change the answer.

Q197

How do you distinguish between AWS Config and CloudTrail in exam questions?

▣ ANSWER

CloudTrail: who did what, when (API call logs, user actions, governance and compliance auditing of actions). AWS Config: what is the current and historical state of resources (configuration changes, compliance checking, resource relationships). Config answers 'what changed in this resource?' CloudTrail answers 'who made the API call?'

Q198

What are AWS global services (not region-specific)?

▣ ANSWER

Global services: IAM (users, roles, policies), Route 53, CloudFront, WAF and Shield (deployed globally but configured per distribution/ALB/API GW), AWS Organizations, Control Tower, Trusted Advisor, AWS Support. These services don't require a region selection when accessed via console.

Q199

How do you handle 'which architecture is MOST resilient' questions?

▣ ANSWER

Look for: Multi-AZ or multi-region deployment, no single points of failure, Auto Scaling groups, managed services with built-in redundancy, Route 53 health checks and failover, decoupled components (SQS, SNS), Circuit breakers (App Mesh), Regular backup and tested restore procedures (AWS Backup, tested in FIS).

Q200

What final revision tips help pass SAP-C02?

▣ ANSWER

Study AWS whitepapers (Well-Architected Framework, Disaster Recovery, Security Best Practices). Practice 20 scenario questions daily. Focus on understanding WHY an answer is correct, not just what. Pay attention to keywords like 'real-time', 'near real-time', 'batch', 'minimize cost', 'no code changes'. Review AWS blogs and re:Invent sessions for recently launched services.

Section 21: Advanced Compute & HPC

Q201-Q210

Q201

What is AWS Graviton and why should you consider it?

▣ ANSWER

AWS Graviton processors (ARM-based, designed by AWS) deliver up to 40% better price/performance than x86 for many workloads. Graviton3: most recent, used in C7g, M7g, R7g instances. Supports containers, Lambda (arm64), RDS, ElastiCache. Ideal for web servers, microservices, databases, video encoding.

Q202

What are EC2 Nitro Instances?

▣ ANSWER

Nitro is the hypervisor platform for modern EC2 instances. Offloads VPC, EBS, instance storage functions to dedicated hardware (Nitro Cards), leaving nearly all host resources for the instance. Enables higher network performance (100 Gbps), bare-metal instances (no hypervisor overhead), and enhanced security via hardware-enforced isolation.

Q203

What is HPC on AWS and which services support it?

▣ ANSWER

HPC uses AWS for computationally intensive workloads. Key services: HPC-optimized instances (Hpc7g, P4d, Trn1), Elastic Fabric Adapter (EFA) for low-latency MPI networking, AWS ParallelCluster (automated HPC cluster management), FSx for Lustre (parallel file system), and AWS Batch for job scheduling.

Q204

What is Amazon EC2 Image Builder?

▣ ANSWER

Image Builder automates creating, maintaining, validating, and distributing EC2 AMIs and container images. Defines image pipelines with: base image, build components (install/configure software), test components (validate), distribution settings (AMI regions, ECR), and schedule. Reduces manual AMI maintenance overhead.

Q205

What are Savings Plans vs Reserved Instances trade-offs?

▣ ANSWER

Savings Plans: flexible (any instance family, region, OS for Compute SP; specific family/region for EC2 SP), covers Lambda and Fargate, commitment in \$/hr. RIs: specific instance type/region/OS/tenancy, convertible RIs allow exchange, Scheduled RIs available, can be bought/sold in Marketplace. RIs better for very predictable, specific workloads.

Q206

What is AWS Elastic Fabric Adapter (EFA)?

▣ ANSWER

EFA is a network interface for EC2 instances that provides OS bypass capabilities for HPC and ML applications. Enables direct memory access (RDMA) for inter-node communication in MPI-based HPC workloads, reducing latency and increasing throughput. Supported on C5n, P4d, Inf1, Hpc instances.

Q207

What is AWS Inferentia and Trainium?

▣ ANSWER

Inferentia (Inf1, Inf2 instances) is AWS-designed ML inference chip delivering up to 3x higher throughput and 40% lower cost vs GPU-based inference. Trainium (Trn1 instances) is purpose-built for deep learning training, delivering best performance-per-cost for training transformer models. Both use AWS Neuron SDK.

Q208

What is AWS Parallel Computing Service?

▣ ANSWER

AWS Parallel Computing Service (PCS) is a managed HPC cluster service (launched 2024) that simplifies running HPC workloads. Manages compute, networking (EFA), storage (FSx for Lustre), and Slurm job scheduler. Auto-scales cluster nodes. Eliminates manual ParallelCluster management for traditional HPC use cases.

Q209

Explain EC2 Fleet vs Spot Fleet.

▣ ANSWER

EC2 Fleet launches and manages a mix of On-Demand, Reserved, and Spot Instances across instance types/AZs using a single API call. Supports allocation strategies: lowest-price, capacity-optimized, price-capacity-optimized. Spot Fleet is specifically for Spot Instances only. EC2 Fleet is the newer, recommended option covering all purchase types.

Q210

What is AWS SimSpace Weaver?

▣ ANSWER

SimSpace Weaver is a managed service for running large-scale spatial simulations (city-scale, logistics, defense). Distributes simulation workload across many EC2 instances. Uses a shared virtual world model. Use for urban planning simulations, logistics optimization, crowd simulation, and training ML models on synthetic data.

Q211

What is AWS IoT Core?

▣ ANSWER

IoT Core is a managed message broker for IoT devices. Supports MQTT, MQTT over WebSocket, HTTPS, LoRaWAN. Device Shadow maintains last-known state. Rules Engine routes messages to 20+ AWS services. Device Registry tracks device metadata. Fleet Hub provides web app for fleet management. Scales to billions of devices.

Q212

What is AWS IoT Greengrass?

▣ ANSWER

Greengrass extends AWS to edge devices, allowing them to run Lambda functions, containers, ML inference (with pre-built ML components), and sync device state locally even without cloud connectivity. Greengrass Core runs on a local gateway. Communicates using MQTT locally and syncs with IoT Core when connected.

Q213

What is AWS IoT TwinMaker?

▣ ANSWER

IoT TwinMaker builds digital twins of real-world systems (factories, buildings, industrial equipment) by connecting data from IoT sensors, cameras, enterprise systems, and 3D models into a unified virtual representation. Provides Grafana plugin for visualization. Use for predictive maintenance, facility management, industrial operations optimization.

Q214

What is AWS Panorama?

▣ ANSWER

Panorama is a computer vision service bringing ML inference to on-premises cameras without sending video to the cloud. A Panorama Appliance (hardware) or Panorama-enabled cameras run CV models locally, streaming only metadata/results. Use for quality inspection, safety monitoring, and people counting on factory floors.

Q215

What is the difference between AWS Local Zones and Wavelength Zones?

▣ ANSWER

Local Zones extend AWS infrastructure to metro areas for ultra-low latency to end users (gaming, media production, ML inference near users). Wavelength Zones embed compute inside telecom carrier 5G networks for mobile/5G applications needing sub-10ms latency. Both are extensions of parent AWS Regions.

Q216

What is AWS Robomaker?

▣ ANSWER

RoboMaker provides a cloud-based simulation service for developing, testing, and deploying robotic applications. Built on Robot Operating System (ROS). Supports simulation world generation (artificial environments), parallel simulation runs, fleet deployment and management. Integrates with IoT Greengrass for edge deployment.

Q217

What is the AWS IoT Device Defender?

▣ ANSWER

IoT Device Defender continuously audits IoT configurations, monitors device behavior (using ML), and sends alerts for deviations. Audit checks: overly permissive policies, revoked certificates in use, IoT logging disabled. Detect: unusual MQTT message volumes, spikes in connections, unexpected communication partners.

Q218

How does LoRaWAN work with AWS IoT Core?

▣ ANSWER

IoT Core for LoRaWAN lets you connect long-range, low-power LoRaWAN devices directly to AWS without building a LoRaWAN Network Server (LNS). Manage gateways, devices, and profiles through IoT Core. Messages flow: LoRaWAN device → LoRa Gateway → IoT Core LNS → IoT Rules Engine → downstream services.

Q219

What is AWS IoT FleetWise?

▣ ANSWER

FleetWise collects, transforms, and transfers vehicle data to the cloud at scale. Uses an intelligent data collection engine deployed on vehicle ECUs to selectively collect data based on campaigns and conditions (only send data when temperature exceeds threshold). Stores data in Timestream or S3. Use for connected vehicle analytics, OTA updates.

Q220

What is AWS IoT SiteWise?

▣ ANSWER

SiteWise collects, organizes, and analyzes industrial equipment data at scale. SiteWise Edge processes data on-premises (gateway). Asset models define equipment hierarchy. Metrics computed at the edge or cloud. SiteWise Monitor provides no-code dashboards for operations teams. Integrates with IoT TwinMaker for digital twins.

Section 23: Governance, Compliance & Audit

Q221-Q230

Q221

What is AWS Audit Manager?

▣ ANSWER

Audit Manager continuously collects evidence of AWS resource configurations and activity to simplify compliance assessments for frameworks like PCI DSS, HIPAA, SOC 2, GDPR, FedRAMP, and CIS Benchmarks. Pre-built and custom frameworks map controls to AWS Config rules, CloudTrail, Security Hub findings, and API calls.

Q222

How does AWS handle GDPR compliance?

▣ ANSWER

AWS provides GDPR-compliant infrastructure: data residency controls (choose Regions), encryption (KMS, TLS), access controls (IAM), audit trails (CloudTrail), data processing agreements (AWS DPA), sub-processor transparency, and data deletion tools (S3 object expiry, RDS deletion, IAM user removal). Customer is responsible for application-level compliance.

Q223

What is AWS Artifact?

▣ ANSWER

Artifact provides on-demand access to AWS security and compliance reports (SOC 1/2/3, PCI DSS AOC, ISO 27001/27017/27018, HIPAA, FedRAMP) and AWS agreements (BAA for HIPAA, GDPR DPA, NDA). Useful for auditors requiring evidence of AWS compliance posture. Downloads available instantly, 24/7.

Q224

What is the difference between SOC 1, SOC 2, and SOC 3?

▣ ANSWER

SOC 1: Controls relevant to financial reporting (SSAE 18). SOC 2: Controls for security, availability, processing integrity, confidentiality, privacy (Trust Service Criteria). SOC 3: Public-facing summary of SOC 2 (no detailed testing). SOC Type 1: point-in-time assessment. SOC Type 2: effectiveness over a period (6+ months, more valuable for enterprises).

Q225

What is AWS License Manager?

▣ ANSWER

License Manager tracks and manages software licenses (Microsoft Windows Server, SQL Server, Oracle, SAP) across AWS and on-premises. Enforces license rules to prevent violations. Associates licenses with AMIs. Tracks consumption on EC2, RDS. Manages Dedicated Host allocations for BYOL. Integrates with AWS Organizations for central management.

Q226

What is AWS Service Catalog?

▣ ANSWER

Service Catalog lets administrators create and manage approved IT service portfolios (CloudFormation stacks) for end users. Portfolios contain products with constraints (launch role, tag, notification). End users deploy standardized, approved products without needing broad IAM permissions. Useful for self-service while maintaining governance.

Q227

What is AWS Config Conformance Pack?

▣ ANSWER

A Conformance Pack is a collection of AWS Config rules and remediation actions packaged as a single unit. Pre-built packs for PCI DSS, HIPAA, CIS Level 1/2, NIST 800-53. Deploy to individual accounts or across an Organization via Organizations integration. Track compliance scores per pack across accounts.

Q228

How does AWS CloudFormation StackSets support governance?

▣ ANSWER

StackSets deploy CloudFormation stacks to multiple accounts and regions simultaneously. With Organizations integration, use service-managed StackSets to auto-deploy to new accounts. Use for: enabling GuardDuty, Config, CloudTrail in all accounts; creating baseline VPCs; deploying security baselines automatically across the entire organization.

Q229

What is AWS Resource Access Manager (RAM)?

▣ ANSWER

RAM shares AWS resources across accounts within an AWS Organization or specific accounts. Shareable resources include: VPC subnets, Transit Gateways, Route 53 Resolver rules, License Manager configurations, Aurora DB Clusters, EC2 Dedicated Hosts, Capacity Reservations. Reduces resource duplication costs.

Q230

What is the difference between AWS Organizations OUs and SCPs?

▣ ANSWER

OUs (Organizational Units) are groups of accounts in a hierarchy for management and policy application. SCPs (Service Control Policies) are permission guardrails applied to OUs/accounts; they define maximum allowed permissions. SCPs don't grant permissions themselves — they restrict what IAM policies can allow within an account.

Section 24: Specialized Storage Services

Q231-Q240

Q231

What is Amazon FSx and what variants are available?

▣ ANSWER

FSx provides fully managed file systems. FSx for Windows File Server: SMB-based, Active Directory integration, NTFS, DFS namespaces. FSx for Lustre: high-performance parallel file system for HPC/ML, integrates with S3. FSx for NetApp ONTAP: multi-protocol NFS/SMB/iSCSI, SnapMirror, FabricPool. FSx for OpenZFS: high-performance NFS.

Q232

What is Amazon S3 Express One Zone?

▣ ANSWER

S3 Express One Zone is a high-performance S3 storage class designed for single-AZ use cases needing consistent single-digit millisecond data access latency. 10x lower latency than S3 Standard. Co-located with compute in specific AZs. Best for: ML training data, financial analytics, media processing, interactive queries with Athena.

Q233

When should you use EFS over S3?

▣ ANSWER

EFS when you need POSIX-compliant shared file system accessible by multiple EC2/ECS/EKS instances simultaneously (concurrent read/write), file locking, hierarchical directory structure, or POSIX permissions. S3 for object storage, web hosting, backup, data lake — not shared mutable file access.

Q234

What is AWS Elastic Disaster Recovery (DRS)?

▣ ANSWER

DRS (formerly CloudEndure DR) provides fast, reliable DR for physical, virtual, and cloud servers. Continuously replicates servers to AWS (sub-second RPO). In case of disaster, launch recovered instances in minutes (RTO in minutes). Differs from MGN (migration tool): DRS is for ongoing DR, not one-time migration.

Q235

What is Amazon S3 Mountpoint?

▣ ANSWER

S3 Mountpoint is an open-source file client that mounts S3 buckets as local file systems on Linux (via FUSE). Supports high-throughput sequential read for ML training, read-only or write-new-objects workloads. Not suitable for random writes or POSIX semantics. Allows existing applications to read S3 data without code changes.

Q236

How does Amazon EFS performance mode work?

▣ ANSWER

EFS General Purpose mode: low latency, suitable for web serving, content management, home directories, general file serving. Max I/O mode: higher aggregate throughput and IOPS, slightly higher latency, ideal for highly parallelized big data, media processing. EFS Throughput modes: Elastic (automatically scales), Bursting (based on stored data size), Provisioned (fixed).

Q237

What is AWS Backup Vault Lock?

▣ ANSWER

Vault Lock enforces WORM compliance on backup vaults. In Compliance mode, no one (including root) can delete vault or change Vault Lock policy once locked. In Governance mode, users with special IAM permissions can delete vaults before lock expires. Meets immutability requirements for HIPAA, FedRAMP, SEC 17a-4.

Q238

What is S3 Batch Operations?

▣ ANSWER

S3 Batch Operations performs large-scale batch operations on S3 objects: copy, invoke Lambda, restore Glacier objects, set ACLs, set tags, enable replication, replace all ACLs with bucket owner ACLs. Creates a job with an inventory manifest (S3 Inventory report). Tracks progress, generates completion report. Processes billions of objects.

Q239

What is Amazon S3 Object Lambda?

▣ ANSWER

S3 Object Lambda intercepts S3 GET, LIST, or HEAD requests and applies a Lambda function to transform data before returning it to the requester. Use cases: data masking/redacting PII, format conversion (XML to JSON), image resizing, custom authentication, serving different content versions without storing multiple copies.

Q240

How does EBS Multi-Attach work?

▣ ANSWER

EBS Multi-Attach allows a single io1 or io2 EBS volume to be attached to up to 16 EC2 instances in the same AZ simultaneously. Applications must manage concurrent write access (cluster-aware file system like OCFS2 or GFS2, or distributed lock manager). Use for high-availability clustered databases or shared storage for cluster applications.

Section 25: SAP-C02 Mock Scenario Review

Q241-Q250

Q241

A startup needs to deploy a mobile backend with unpredictable traffic. Most cost-effective architecture?

▣ ANSWER

Use API Gateway (HTTP API) + Lambda for backend logic + DynamoDB On-Demand for database. All services scale automatically from zero to peak traffic. Pay only for actual usage. No idle resources. Use Cognito for authentication. S3 + CloudFront for static assets. Total cost near-zero during low traffic, scales automatically during spikes.

Q242

A company has a legacy app that cannot be modified but needs to scale. What is the best approach?

▣ ANSWER

Rehost (lift-and-shift) to EC2 using Application Migration Service. Put ALB in front. Add Auto Scaling Group for horizontal scaling. Migrate database to RDS (same engine, minimal changes). Enable CloudFront for caching. Use ElastiCache for session management if app is stateless-capable. Monitor with CloudWatch and optimize over time.

Q243

How would you implement a disaster recovery plan for an RTO of 15 minutes and RPO of 5 minutes?

▣ ANSWER

Warm Standby strategy in a secondary region. Continuous database replication: Aurora Global Database (sub-second replication, RPO ~1 second) or RDS with cross-region read replica (RPO ~5 minutes). Reduced-capacity application tier in secondary region (smaller ASG). Route 53 health-check-based failover. Regular DR drills using FIS.

Q244

A company wants to minimize operational overhead for running containers. Which service?

▣ ANSWER

AWS Fargate with ECS or EKS. No EC2 instances to manage, patch, or right-size. Define CPU/memory per task. Integrates with ALB, CloudWatch, X-Ray, IAM task roles, Secrets Manager. For Kubernetes-specific requirements or multi-cloud portability, use EKS Fargate. For pure AWS-native, ECS Fargate is simpler and cheaper.

Q245

An e-commerce app needs to handle flash sale traffic (100x normal). What architecture handles this?

▣ ANSWER

CloudFront + S3 for static content (product catalog, images). API Gateway + Lambda for dynamic requests (auto-scales). DynamoDB On-Demand for cart/orders. SQS to decouple order processing. Pre-warm Lambda concurrency with Provisioned Concurrency if cold starts are a concern. Test with AWS Load Testing or Distributed Load Testing Solution.

Q246

A company processes sensitive financial data and needs to meet PCI DSS. Key architectural requirements?

▣ ANSWER

Network isolation: VPC with private subnets, no public internet access to data tier, NACLs + SGs. Data encryption: KMS for data at rest, TLS 1.2+ in transit, no unencrypted protocols. Access control: MFA, least privilege IAM, no shared credentials. Audit: CloudTrail, Config, VPC Flow Logs. WAF + Shield for web-facing components. Annual PCI audit using Audit Manager.

Q247

How do you migrate a 100 TB on-premises database to AWS with less than 1-hour downtime?

▣ ANSWER

Phase 1: Use DMS full-load task (can take days/weeks) with production database running. Phase 2: Enable CDC in DMS to replicate ongoing changes. Phase 3: Monitor DMS latency until near-zero lag. Phase 4: During maintenance window, stop application, allow DMS to sync final changes, redirect application to AWS database. Total downtime: minutes.

Q248

A media company needs to transcode 10,000 videos simultaneously. Best approach?

▣ ANSWER

Use AWS Elemental MediaConvert for managed video transcoding (pay-per-minute, scales automatically, no infrastructure). For custom transcoding, use EC2 Spot Instances with Auto Scaling + SQS job queue (Spot reduces cost 90%). Store inputs in S3, outputs in S3/CloudFront. ECS Fargate Spot is also viable for containerized ffmpeg workloads.

Q249

What is the most secure way to give a third-party vendor temporary access to a specific S3 bucket?

▣ ANSWER

Create an IAM role in your account with a policy restricted to the specific S3 bucket/prefix and actions needed. Add a trust policy allowing the vendor's AWS account to assume the role. Vendor uses sts:AssumeRole to get temporary credentials (max 12 hours). Alternatively, generate pre-signed URLs for specific objects without sharing credentials.

Q250

A company needs real-time fraud detection as transactions are processed. What architecture?

▣ ANSWER

Transaction → API Gateway → Lambda (enrich + validate) → Kinesis Data Streams → Kinesis Data Analytics with ML model (Flink + SageMaker endpoint) → real-time decision (approve/flag/reject) returned via SQS. Flagged transactions → SNS alert → security team. Historical data → S3 → SageMaker for model retraining. Sub-100ms latency achievable.

Q251

What are CloudFront Origin Access Control (OAC) and Origin Access Identity (OAI)?

▣ ANSWER

OAI (legacy): special CloudFront identity used in S3 bucket policies to allow only CloudFront to access S3. OAC (recommended): newer, supports S3 SSE-KMS, all S3 regions, PUT/DELETE operations, server-side encryption with customer keys (SSE-C). Use OAC for new deployments; migrate OAI to OAC.

Q252

What is CloudFront Origin Shield?

▣ ANSWER

Origin Shield is an additional caching layer between CloudFront edge and origin. Acts as a central cache, reducing origin load by collapsing cache misses from all edge locations into a single origin request. Reduces origin traffic up to 73%, improves cache hit ratio, provides better performance for origins in one region.

Q253

What is CloudFront Field-Level Encryption?

▣ ANSWER

Field-Level Encryption adds an extra layer of security beyond HTTPS. Specific data fields (e.g., credit card numbers) are encrypted at the CloudFront edge using a public key. Only the application server with the private key can decrypt the data. Even AWS employees and ALB cannot see the decrypted values.

Q254

How does Lambda@Edge support A/B testing?

▣ ANSWER

Lambda@Edge on Viewer Request reads/sets cookies to assign users to A or B group. Modifies the Origin Request to forward to different origin (different S3 path or different backend). Persists user assignment in cookie. Allows gradually rolling out new features to a percentage of users without DNS changes or multiple CloudFront distributions.

Q255

What is AWS Global Accelerator vs Route 53 failover?

▣ ANSWER

Global Accelerator: static Anycast IPs, instant failover (<30 seconds), traffic routed over AWS backbone, supports TCP/UDP non-HTTP. Route 53 Failover: DNS-based, TTL-dependent failover (typically 60+ seconds), HTTP/HTTPS, no static IPs. GA for lower failover time and non-HTTP; Route 53 for HTTP with acceptable failover time.

Section 27: Cost Optimization Advanced

Q256-Q260

Q256

What is EC2 Spot Instance Interruption handling best practice?

▣ ANSWER

Monitor instance interruption notices (2-minute warning) via EC2 metadata (latest/meta-data/spot/termination-time) or EventBridge. On notice: checkpoint state to S3/DynamoDB, deregister from ELB, drain connections gracefully, and terminate. Use Spot with diversification across multiple instance families/AZs to reduce interruption probability.

Q257

How do you use AWS Cost Anomaly Detection in a multi-account org?

▣ ANSWER

Create an AWS Cost Monitor at Organization, Linked Account, Service, or Cost Allocation Tag level. Set alert thresholds (% above expected or absolute \$ amount). Subscribe an SNS topic for notifications. Investigate anomalies in Cost Explorer. Common causes: forgotten resources, configuration change that increased API calls, unexpected data transfer.

Q258

What is AWS Compute Optimizer and how does it work?

▣ ANSWER

Compute Optimizer analyzes 14 days of CloudWatch utilization metrics using ML to provide right-sizing recommendations for EC2 instances, ECS on Fargate, Lambda functions, EBS volumes, and Auto Scaling Groups. Shows current vs. recommended with projected cost savings and performance improvement. Integrates with Organizations for org-wide recommendations.

Q259

How do you manage cost in a multi-tenant SaaS architecture on AWS?

▣ ANSWER

Tag resources per tenant. Use AWS Cost Allocation Tags activated in Cost Explorer. Use separate AWS accounts per tenant tier (large tenants) for isolation and cost tracking. Use AWS Cost and Usage Report (CUR) + Athena + QuickSight for per-tenant cost dashboards. Implement charge-back/show-back reporting. Use DynamoDB On-Demand for variable tenant usage.

Q260

What is AWS Free Tier and how does it affect architecture decisions?

▣ ANSWER

Free Tier offers 12-month free (750 hrs t2/t3.micro, 5 GB S3, 25 GB DynamoDB, etc.), always-free (1M Lambda requests/month, 25 GB DynamoDB, CloudWatch 10 metrics), and trial offers. For new accounts, design within free tier limits for development. Note: Free Tier does not apply to data transfer out in all cases.

Section 28: Advanced IAM & Federation

Q261-Q265

Q261

What is AWS IAM Roles Anywhere?

▣ ANSWER

IAM Roles Anywhere allows workloads outside AWS (on-premises servers, VMs, containers) to obtain temporary AWS credentials using X.509 digital certificates issued by a trusted Certificate Authority (CA). Eliminates long-term IAM user credentials for on-premises systems. Configured with trust anchors and profiles.

Q262

What is the difference between Resource-based policies and Identity-based policies for cross-account?

▣ ANSWER

For cross-account access to resources like S3: resource-based policy (S3 bucket policy) allows the external account — no need for the principal to also have identity-based policy allowing s3:GetObject (resource-based policy alone is sufficient for cross-account). For services that don't support resource-based policies (like EC2), must use IAM role assumption.

Q263

What is AWS Managed Microsoft AD?

▣ ANSWER

AWS Managed Microsoft AD is a fully managed Active Directory in AWS (Windows Server AD). Supports trusts with on-premises AD, Kerberos, LDAP, Group Policy. Enables AD-integrated workloads (SharePoint, RDS for SQL Server, Windows EC2 domain join) without managing AD infrastructure. Can extend on-premises AD to AWS via trust relationship.

Q264

What is an IAM Service-Linked Role?

▣ ANSWER

A service-linked role is an IAM role predefined by and linked to a specific AWS service. The service assumes this role to perform actions on your behalf. You cannot edit the trust policy (linked to the specific service). You can only delete if no resources use the service. Example: `AWSServiceRoleForAutoScaling`, `AWSServiceRoleForECS`.

Q265

How does multi-factor authentication (MFA) work in AWS IAM?

▣ ANSWER

MFA adds a second factor to IAM authentication. Supported devices: virtual MFA app (Google Authenticator, Authy), FIDO security key (YubiKey), hardware TOTP token. Enable per IAM user in console. Enforce MFA using IAM condition key `aws:MultiFactorAuthPresent` in policies. SCP can enforce MFA for console/API access across an organization.

Section 29: AWS Well-Architected Tool & Trusted Advisor

Q266–Q270

Q266

How do you use AWS Trusted Advisor programmatically?

▣ ANSWER

Use AWS Support API (requires Business or Enterprise Support plan) to call `DescribeTrustedAdvisorChecks`, `DescribeTrustedAdvisorCheckResult`, `RefreshTrustedAdvisorCheck`. Integrate with EventBridge for automated alerts on check status changes. Use Lambda + CloudWatch to auto-remediate issues like open S3 buckets or exposed IAM access keys.

Q267

What are Trusted Advisor's service limit checks?

▣ ANSWER

Trusted Advisor monitors service quotas (EC2 instances per region, VPCs, Elastic IPs, etc.) and alerts when usage reaches 80% of the limit. Integrate with EventBridge → Lambda to automatically request quota increases via Service Quotas API before limits are hit. Use Service Quotas console for central quota management across services.

Q268

What is AWS Service Quotas?

▣ ANSWER

Service Quotas provides a centralized view of quotas (limits) for AWS services. Request quota increases for adjustable quotas directly from the console or API. Set CloudWatch alarms when approaching quota limits. Integrates with Organizations to manage quotas across accounts. Replaces the fragmented approach of contacting support per service.

Q269

How do you interpret Well-Architected Tool risk levels?

▣ ANSWER

HRI (High Risk Issue): significant vulnerability that may cause failure, security breach, or major cost issue. MRI (Medium Risk Issue): less severe but should be addressed. No Risk: best practice followed. Improvement Plan prioritizes HRIs first. Schedule regular reviews (quarterly recommended) and track improvement over milestones.

Q270

What is AWS re:Post and when is it useful for SAP-C02 preparation?

▣ ANSWER

re:Post is AWS's Q&A; community (replaced AWS Forums). Architects, developers, and AWS staff answer technical questions. Useful for finding answers to specific architectural edge cases, understanding service limitations, and finding community-tested solutions. Complements official AWS documentation for practical implementation guidance.

Q271

What are the key differences between SAA-C03 and SAP-C02 exam depth?

▣ ANSWER

SAP-C02 goes deeper on: multi-account governance (Organizations, Control Tower, SCPs), hybrid connectivity (Direct Connect, VPN architectures, TGW), migration strategies (DMS, MGN, Snow family), advanced security architectures, cost optimization at scale, Well-Architected Framework application, and complex multi-tier scenarios with multiple valid answers requiring trade-off analysis.

Q272

What AWS whitepapers are most important for SAP-C02?

▣ ANSWER

Essential whitepapers: AWS Well-Architected Framework (and all 6 pillar whitepapers), Disaster Recovery whitepaper, AWS Security Best Practices, Overview of Amazon Web Services, Architecting for the Cloud (AWS Best Practices), AWS Storage Services Overview, Running Containerized Microservices on AWS, and Microservices on AWS.

Q273

How should you approach cost optimization questions with multiple valid answers?

▣ ANSWER

Evaluate each option for: immediate cost reduction vs. long-term savings, operational overhead introduced, risk to availability, and whether the solution actually addresses the stated workload pattern. Savings Plans > Reserved Instances for flexibility. Spot for stateless/batch. Managed services reduce operational overhead (which has a cost). Architect for minimum waste.

Q274

What is the recommended study timeline for SAP-C02?

▣ ANSWER

Recommended: 3-4 months for SAA-C03 holders, 6 months for SAA-C03 aspirants. Week 1-4: Core services deep dive. Week 5-8: Advanced networking, security, multi-account. Week 9-12: Migration, DR, cost optimization. Week 13-16: Practice exams, review weak areas, read AWS whitepapers. Take 3-4 full practice exams before sitting the real exam.

Q275

What are the exam format details for AWS SAP-C02?

▣ ANSWER

SAP-C02 has 75 questions (65 scored + 10 unscored) in 180 minutes. Question types: single-answer multiple choice (1 correct of 4 options) and multiple-response (2-3 correct of 5 options). Passing score: 750/1000. Available in 10+ languages. Scenario-based questions with 4-6 sentence stems. Cost: \$300 USD. Certification valid for 3 years.

Section 31: AWS Organizations & Landing Zone

Q276-Q278

Q276

What is AWS Control Tower's Account Factory?

▣ ANSWER

Account Factory automates provisioning of new AWS accounts with pre-approved baselines: VPC configuration, CloudTrail, AWS Config, IAM Identity Center permission sets, and account email. Uses AWS Service Catalog portfolio. New accounts follow organizational standards automatically. Account Factory for Terraform (AFT) enables GitOps-based account vending.

Q277

What are Guardrails in AWS Control Tower?

▣ ANSWER

Guardrails are pre-packaged governance rules. Preventive guardrails use SCPs to prevent actions (e.g., deny deletion of CloudTrail logs). Detective guardrails use AWS Config Rules to detect non-compliant configurations (e.g., detect if CloudTrail is disabled). Elective guardrails are optional. Over 400 built-in guardrails across security, operations, and compliance.

Q278

What is AWS Organizations Tag Policies?

▣ ANSWER

Tag Policies enforce consistent tagging across accounts in an AWS Organization. Define required tag keys, allowed values, and case requirements. Non-compliant resources show as non-compliant in the Tag Policy compliance dashboard. Use with Config Rules for automated remediation. Helps maintain consistent cost allocation and resource management.

Q279

What is SQS visibility timeout and how do you tune it?

▣ ANSWER

Visibility timeout is the period a message is hidden from other consumers after being received. Default: 30 seconds. Should be set longer than your processing time. If processing takes 5 minutes, set to 6+ minutes. Use `ChangeMessageVisibility` API to extend if needed. If Lambda processes SQS, Lambda automatically sets visibility timeout to 6x the function timeout (max 900 seconds).

Q280

What is SNS message filtering?

▣ ANSWER

SNS filter policies (JSON) applied to subscriptions define conditions on message attributes. Only messages matching the filter policy are delivered to that subscriber. Reduces unnecessary SQS/Lambda invocations. Example: route 'type: order_placed' to order processor and 'type: payment_failed' to billing service from a single SNS topic.

Q281

What is the difference between Kinesis Enhanced Fan-Out and standard GetRecords?

▣ ANSWER

Standard GetRecords: polling model, 2 MB/s total per shard shared across all consumers, higher latency (up to 200ms). Enhanced Fan-Out: push model via HTTP/2, dedicated 2 MB/s throughput per registered consumer per shard, ~70ms propagation latency. Use Enhanced Fan-Out for latency-sensitive, multiple-consumer scenarios.

Q282

What is RDS Proxy and when should you use it?

▣ ANSWER

RDS Proxy maintains a pool of established database connections, reducing connection overhead for serverless or highly concurrent applications (Lambda functions). Reduces failover time by 66% (connections maintained during failover). Supports IAM authentication and Secrets Manager integration. Supports MySQL, PostgreSQL, MariaDB, SQL Server, Aurora.

Q283

What is Aurora Serverless v2?

▣ ANSWER

Aurora Serverless v2 auto-scales Aurora capacity in fine-grained increments (0.5 ACU steps) in milliseconds based on actual workload. Scales from 0.5 to 128 ACUs. Supports Multi-AZ, read replicas, Global Database. Pay per ACU-second. Ideal for variable workloads (dev/test, SaaS with per-tenant databases, infrequently used applications).

Q284

What is Amazon Timestream?

▣ ANSWER

Timestream is a serverless time-series database for IoT and operational analytics. Automatically scales and manages storage tiers (in-memory for recent data, SSD for recent history, S3 for historical). Built-in time-series analytics functions. 1,000x faster and 1/10 the cost of relational DB for time-series. Integrates with Grafana, Kinesis, Lambda, IoT Core.

Section 34: API & Microservices

Q285–Q286

Q285

What is the difference between REST API and HTTP API in API Gateway?

▣ ANSWER

HTTP API: lower latency, lower cost (70% cheaper), supports JWT authorizers, OIDC, Lambda proxy, HTTP proxy, CORS, no API keys or resource policies. REST API: full-featured, request/response transformations, API keys, usage plans, WAF integration, caching (custom TTL), resource policies, X-Ray tracing, mock integrations, older but more features.

Q286

What is service discovery in AWS for microservices?

▣ ANSWER

AWS Cloud Map provides service discovery. Register services with DNS names or API. ECS integrates with Cloud Map automatically (service discovery via Route 53 private hosted zones). EKS uses CoreDNS + Kubernetes Services. App Mesh uses Envoy to discover services. Alternative: ALB + ECS service with ALB-based routing per service path.

Section 35: Backup & Recovery

Q287-Q288

Q287

How do you test disaster recovery on AWS?

▣ ANSWER

Use AWS Fault Injection Simulator (FIS) to simulate AZ/region failures. AWS Resilience Hub validates architecture against RTO/RPO targets. Schedule regular DR drills with documented runbooks. Test: RDS failover, Route 53 DNS failover, cross-region restoration from S3/AWS Backup, ECS service redeployment in secondary region. Document lessons learned.

Q288

What is the 3-2-1 backup rule and how does AWS implement it?

▣ ANSWER

3-2-1: 3 copies of data, on 2 different storage types, with 1 copy offsite. AWS implementation: primary data in RDS Multi-AZ (2 copies, 2 AZs = 1 media type), automated snapshots to S3 (2nd storage type), cross-region snapshot copy (offsite). AWS Backup automates this. Object Lock adds immutability protection.

Section 36: Networking Advanced

Q289-Q290

Q289

What is BGP Communities in AWS Direct Connect?

▣ ANSWER

BGP Communities are tags on route advertisements used to control routing behavior. In Direct Connect: local preference community affects routing between Direct Connect paths; no-export community prevents routes from being advertised to other BGP peers. Use communities to prefer one DX connection over another or to control traffic routing across multiple connections.

Q290

What is AWS Transit Gateway Connect?

▣ ANSWER

Transit Gateway Connect attachments support GRE tunnel + BGP protocol for high-bandwidth connectivity (up to 5 Gbps per attachment using ECMP across multiple GRE tunnels). Use for SD-WAN appliances, third-party network appliances, or any device supporting GRE/BGP. More performant than VPN attachments (1.25 Gbps limit).

Section 37: Monitoring Advanced

Q291-Q292

Q291

What is Amazon CloudWatch Evidently?

▣ ANSWER

Evidently is a feature flag and A/B experimentation service. Define features with multiple variations, assign traffic percentages to each, and measure metrics (custom CloudWatch metrics) to determine which variation performs better. Integrates with CloudWatch for metrics analysis. Allows gradual feature launches with automatic rollback triggers.

Q292

What is CloudWatch Logs Insights?

▣ ANSWER

Logs Insights is an interactive query service for CloudWatch Logs. Uses a purpose-built query language. Commands: fields, filter, stats, sort, limit, parse (regex). Can query multiple log groups simultaneously. Saves queries. Visualizes results as bar charts or time-series. Useful for ad-hoc investigation, creating CloudWatch dashboards from log data.

Q293**How does ElastiCache improve application performance?****ANSWER**

ElastiCache reduces database load by caching frequently accessed data. Lazy loading: cache miss → fetch from DB → write to cache. Write-through: write to cache and DB simultaneously. Use for: user session storage (Redis String), leaderboards (Redis Sorted Sets), real-time analytics (Redis), simple key-value caching (Memcached or Redis). Sub-millisecond response times.

Q294**What is Amazon CloudSearch?****ANSWER**

CloudSearch is a managed search service for websites or applications. Supports 34 languages, full-text search, faceted search, result highlighting, autocomplete, and geospatial search. Automatic scaling. However, AWS recommends migrating to OpenSearch Service for new workloads as it has more features, better performance, and OpenSearch maintains active development.

Q295**How does Route 53 Application Recovery Controller work?****ANSWER**

ARC provides readiness checks (continuously validates replica resources are correctly provisioned) and routing controls (manual or automatic failover switch between regions, like a circuit breaker). Routing controls use Route 53 health checks to drive DNS failover. Provides high confidence in DR capability before actual disaster occurs.

Q296

What is Amazon Route 53 Resolver?

▣ ANSWER

Route 53 Resolver provides DNS resolution for hybrid cloud. Inbound endpoints: DNS queries from on-premises to resolve AWS (VPC) DNS names. Outbound endpoints: EC2 instances in VPC resolve on-premises DNS names via forwarding rules. DNS Firewall: filter DNS queries at the resolver level using domain lists (block/allow malicious domains). Integrates with Firewall Manager.

Section 40: Career & Certification

Q297-Q300

Q297

What is the difference between AWS Professional and Specialty certifications?

▣ ANSWER

Professional certifications (Solutions Architect Professional, DevOps Engineer Professional) validate advanced skills in designing and operating complex AWS environments. Specialty certifications (Advanced Networking, Security, Machine Learning, Database, Data Analytics, SAP on AWS) validate deep expertise in specific technical domains. All are 3-year certifications.

Q298

What does it mean to re-certify an AWS Professional certification?

▣ ANSWER

AWS certifications expire after 3 years. To recertify: pass the current exam (gets another 3-year validity), pass a higher-level exam (passing Professional recertifies all lower-level certs), or pass any specialty exam (recertifies all lower-level certs). AWS also accepts AWS Partner recertification paths through training programs.

Q299

What is the AWS Certification benefits program?

▣ ANSWER

AWS Certified individuals receive: digital badges (shareable on LinkedIn, Acclaim), access to AWS Certification SME programs, discounts on future exam vouchers (50% off), access to exclusive events at AWS re:Invent, AWS Certified LinkedIn community, free practice exams, and early access to new AWS service content.

How do you use AWS Skill Builder for SAP-C02 preparation?

▣ ANSWER

AWS Skill Builder offers: SAP-C02 official exam guide, official practice exam (20 questions), digital course 'Advanced Architecting on AWS', learning plans with role-based paths, and AWS Partner training. Skill Builder Individual subscription (\$29/month) includes enhanced practice exams with explanations. Official practice exams use same question format as real exam.